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Capstone Project Document

THE LAST SIGNAL: A 2D Top-Down Asymmetric Multiplayer Horror Game with Dedicated Game Servers and an Online Service Platform.



Group Name: GFA26SE54	
Group Members	Cao Thành Danh – SE194497 – Team Leader Nguyễn Công Thiên Ân – SE194544 – Member Hồ Trọng – SE194602 – Member Phan Doãn Thuận – SE194684 – Member

Supervisor	Nguyễn Thế Hoàng
Ext Supervisor	N/A
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Definition and Acronyms

The table below lists the acronyms and domain-specific terms used throughout this document.

Acronym	Definition
NGO	Unity Netcode for GameObjects — the multiplayer networking framework used
UTP	Unity Transport Package
UGS	Unity Gaming Services (Relay, Authentication)
CCU	Concurrent Users (Relay service billing metric)
FSM	Finite State Machine
FOV	Field of View — raycast-based enemy vision detection
NavMesh	Navigation Mesh used for AI pathfinding
URP	Universal Render Pipeline
RPC	Remote Procedure Call
LAN	Local Area Network
HUD	Heads-Up Display
FR	Functional Requirement
NFR	Non-Functional Requirement
MVP	Minimum Viable Product
SRS	Software Requirement Specification
SAD	Software Architecture Document
GDD	Game Design Document
UC	Use Case
API	Application Programming Interface
GUI	Graphical User Interface

III. Software Requirement Specification

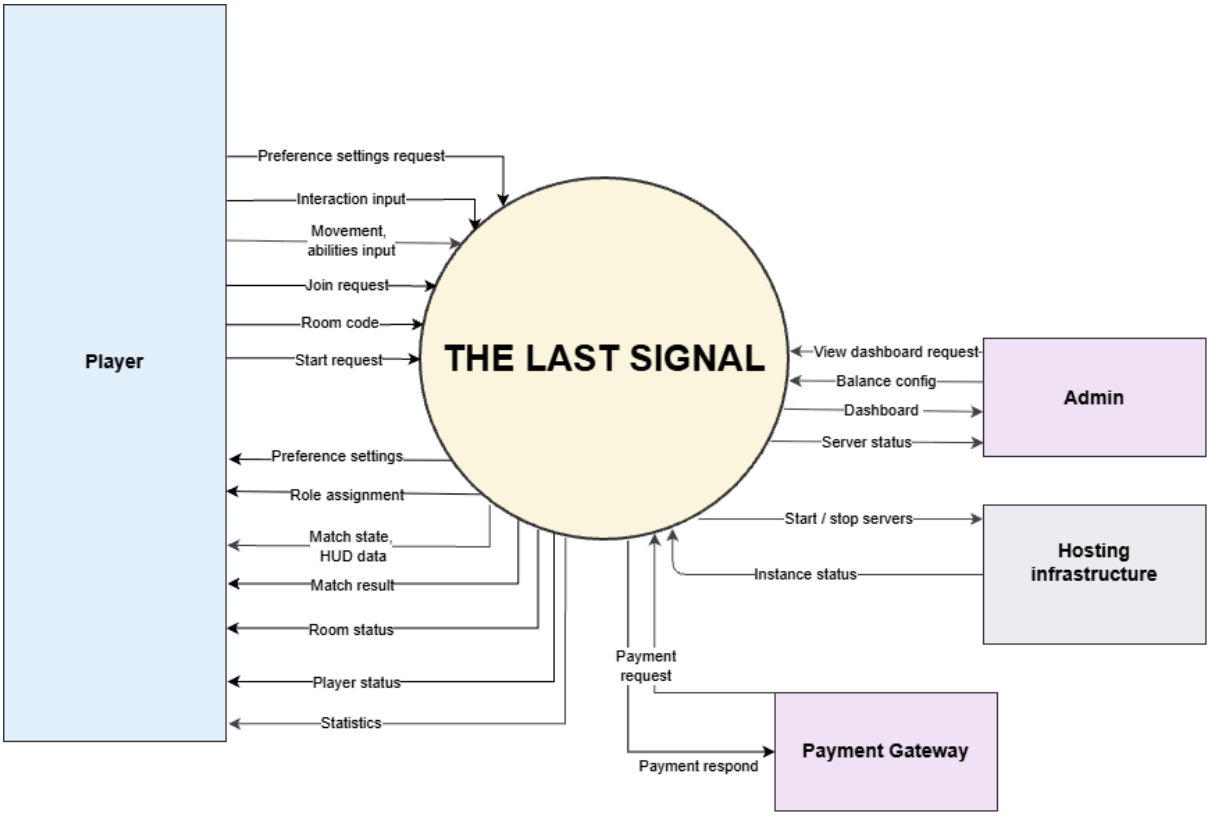
This section specifies the functional and non-functional requirements for THE LAST SIGNAL following the detailed Software Requirement Specification (Report 3) structure. Requirements use MoSCoW prioritisation (Must / Should / Could) so scope can degrade under schedule pressure without abandoning the project's core identity.

1. Product Overview

THE LAST SIGNAL consists of four cooperating software components and integrates with external infrastructure services. The game client is a Unity 2D application for Windows where players authenticate, manage profiles, browse and purchase cosmetic skins, create or join rooms by shareable room code, and play asymmetric 3-versus-1 matches. The dedicated game server is a headless Linux build using Unity Netcode for GameObjects (NGO) and Unity

Transport (UTP) that runs one match instance and is authoritative over all gameplay-critical state. The backend is a Java 21 (Spring Boot) web service with a PostgreSQL database that handles authentication, player profiles, skin store transactions, room lifecycle, dedicated server orchestration, and match result persistence. The admin portal is a React web application used exclusively by administrators to manage players, balance configurations, match records, bug reports, analytical dashboards, and store transactions.

The context diagram below illustrates the boundary of THE LAST SIGNAL and the external entities that interact with it.



2. User Requirements

2.1 Actors

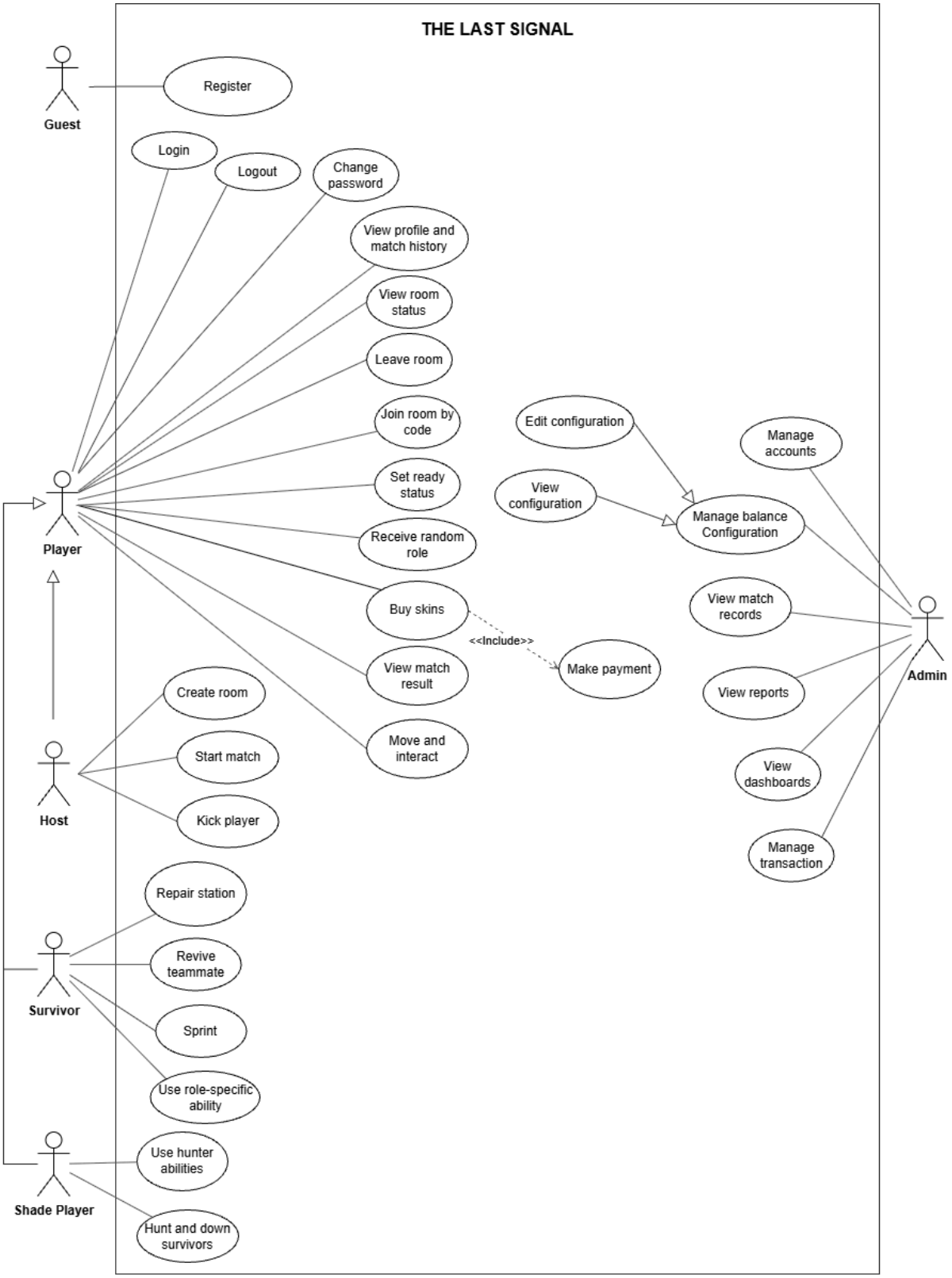
The table below defines the human and external system actors that interact with THE LAST SIGNAL across the Unity Game Client, Dedicated Game Server, Spring Boot Backend, and Web Admin Portal.

#	Actor	Description
1	Guest	An unauthenticated user launching the Unity Game Client who must register a new account or log in with existing credentials before accessing game features.
2	Player	An authenticated user on the Unity Game Client who can manage their account password, view their profile and match history,

		browse and purchase cosmetic skins, create or join a room by code, set ready status, enter a match, receive a random role (Survivor or The Shade), reconnect if disconnected, view match results, and submit bug/issue reports.
3	Room Owner / Host	A specialization of Player who creates a room, receives a shareable 6-character room code, can kick non-host players while the room is waiting, and initiates the match once all four players are ready.
4	Survivor	An in-match specialization of Player randomly assigned to three of the four players at match start (uniquely distributed among Drone Operator, Engineer, and Logistic). Survivors explore the dark station, manage Sanity, Health, and Stamina, repair broken machines via on-screen minigames, use role-specific abilities, revive downed teammates, and escape through unlocked Emergency Exits.
5	The Shade (Shade Player)	An in-match specialization of Player randomly assigned to one of the four players at match start. Controls the Hunter (The Shade), uses Hunter abilities, hunts and downs all Survivors before they can escape, and suffers configured debuffs when exposed to illuminated areas.
6	Admin	An authorized administrator who signs in to the Web Admin Portal to manage player accounts, view and edit versioned game balance configurations, inspect match records, review bugs, monitor analytical dashboards, and audit store transactions.

2.2 Use Cases

2.2.1 Diagram(s)



2.2.2 Descriptions

The table below describes all 31 use cases depicted in the Use Case Diagram, numbered sequentially from UC-01 to UC-31 with their corresponding actors.

ID	Use Case	Actors	Use Case Description
UC-01	Register	Guest	Allows an unauthenticated Guest on the Unity Game Client to create a new player account with a unique username, email, and password.
UC-02	Login	Guest	Allows a Guest to authenticate into the Unity Game Client (or an Admin into the Web Admin Portal) using valid credentials, issuing a JWT session and supporting automatic login on subsequent client launches.
UC-03	Logout	Player, Admin	Allows an authenticated Player or Admin to terminate their current session and clear stored authentication tokens.
UC-04	Change password	Player	Allows an authenticated Player to update their account password after verifying their current password.
UC-05	Create room	Host	Allows an authenticated Player to create a server-hosted multiplayer room as the Host and obtain a unique 6-character room code to share with other players.
UC-06	Join room by code	Player	Allows an authenticated Player to join an existing WAITING room by entering its valid 6-character room code, provided the room has fewer than four players.
UC-07	View room status	Player, Host	Allows players in a room lobby to view the room code, current connected players, Host indicator, and each player's readiness state.
UC-08	Set ready status	Player, Host	Allows a Player in the room lobby to toggle their readiness status (Ready / Not Ready) prior to match initiation.
UC-09	Start match	Host	Allows the Host to request match initiation once all four connected players are Ready. The backend coordinates dedicated game server allocation with the Hosting Infrastructure, assigns random 3v1 roles, and issues per-player join tokens.

UC-10	Kick player	Host	Allows the Host to remove a non-host player from the room lobby while the room is in the WAITING state.
UC-11	Leave room	Player, Host	Allows a Player or Host to leave the room lobby before the match starts. If the Host leaves a WAITING room, Host authority automatically migrates to the next remaining player, or the room closes if empty.
UC-12	Receive random role	Player (Survivor / Shade Player)	Automatically assigns three players to unique Survivor roles (Drone Operator, Engineer, Logistic) and one player to The Shade when the match starts.
UC-13	Move and Interact	Survivor, Shade Player	Allows a player in a match to move in eight directions and interact with eligible environmental objects (such as doors, broken machines, downed teammates, or unlocked Emergency Exits).
UC-14	View profile and match history	Player	Allows an authenticated Player on the Unity Client to view their profile information, aggregated career statistics, and history of completed matches.
UC-15	Buy skins	Player	Allows an authenticated Player on the Unity Client to browse cosmetic skins in the in-game store, initiate a purchase, and unlock the skin upon payment completion (includes UC-16 Make payment).
UC-16	Make payment	Player	Processes the payment transaction with the configured Payment Gateway (Sandbox/Mock) when included by UC-15 Buy skins, verifying transaction status before granting skin ownership.
UC-17	Repair station	Survivor	Allows any Survivor to interact with a broken machine and complete an on-screen repair minigame. The Engineer repairs faster than other roles. Completing a repair illuminates that machine's room (applying a light debuff to The Shade) and increments shared repair progress.
UC-18	Use role-specific ability	Survivor	Allows each Survivor to activate the ability tied to their assigned role: Drone Operator pilots a recon drone (spotlight + motion radar),

			Engineer benefits from faster repair speed and higher Stamina capacity, and Logistic triggers a temporary stun against The Shade.
UC-19	Revive teammate	Survivor	Allows a living Survivor to interact with a Downed teammate over a channel duration to restore them to active play with partial Health.
UC-20	Sprint	Survivor	Allows a Survivor to temporarily boost movement speed while consuming Stamina.
UC-21	Use hunter abilities	Shade Player	Allows the Shade Player to activate The Shade's Hunter-specific tracking or ambush abilities subject to cooldowns and light-exposure rules.
UC-22	Hunt and down Survivors	Shade Player	Allows the Shade Player to pursue and attack Survivors to reduce their Health to zero (Downed), while suffering configured debuffs when exposed to illuminated areas.
UC-23	View match result	Player	Allows players to view the authoritative match outcome (Survivor Victory if at least one Survivor escaped; Shade Victory if all Survivors were downed) and per-player statistics at match conclusion.
UC-24	Manage players	Admin	Allows an Admin on the Web Admin Portal to search player accounts, inspect profiles and statistics, and update account activation status (ACTIVE / DEACTIVE via enabled flag).
UC-25	Manage balance configuration	Admin	Generalized administrative use case for overseeing versioned game balance parameters (specialized into UC-26 View configuration and UC-27 Edit configuration).
UC-26	View configuration	Admin	Allows an Admin to inspect current and historical versions of game balance parameters (repair counts, Sanity drain/thresholds, Stamina rates and Shade light debuffs).
UC-27	Edit configuration	Admin	Allows an Admin to create or update versioned game balance parameter values applied to new dedicated server match instances.
UC-28	View match records	Admin	Allows an Admin to browse and inspect persisted match session records, participant roles, durations, and final outcomes.

UC-29	View reports	Admin	Allows an Admin to review and update the status of bug reports, game issue reports submitted by players.
UC-30	View dashboards	Admin	Allows an Admin to view analytical dashboards displaying platform metrics, active rooms/servers, match completion counts, and Survivor vs. Shade win rates.
UC-31	Manage transaction	Admin	Allows an Admin to view, filter, audit, and reconcile cosmetic skin store purchase transactions and payment statuses.
UC-32	Submit report	Player	Allows an authenticated Player on the Unity Client to submit a bug or gameplay issue report to the backend platform for administrator review.

3. Functional Requirements

3.1 System Functional Overview

THE LAST SIGNAL provides two distinct user-facing applications backed by a shared backend platform and authoritative dedicated game servers: (1) the Unity 2D Game Client for Guests and Players (who become Hosts in the lobby and Survivors or The Shade during a match), and (2) the React Web Admin Portal exclusively for Administrators.

3.1.1 Screens Flow

1. Unity Game Client Screen Flow: Upon launching the executable, an unauthenticated user lands on the Login Screen (with navigation to the Register Screen). If a valid JWT session is stored locally from a previous session, the client performs Auto-Login and transitions directly to the Main Menu Screen. From the Main Menu Screen, the authenticated Player can navigate to: (a) the Room Lobby Screen (by creating a new room or entering a 6-character room code), (b) the Profile & Match History Screen (including Change Password), (c) the Skin Store & Payment Screen, or (d) the Bug / Issue Report Modal. When all four players in the Room Lobby Screen mark Ready and the Host clicks Start Match, all four clients connect to the allocated Dedicated Game Server and transition into the In-Game HUDScreen (with the Repair Minigame Overlay for Survivors and Pause/Reconnect overlays). When a win condition is met, all clients transition to the Match Result Screen and subsequently return to the Main Menu Screen.

2. Web Admin Portal Screen Flow: Administrators open the Web Admin Portal in a browser and authenticate at the Admin Login Screen. Upon verification of the ADMIN role, the administrator enters the Admin Dashboard Screen and can navigate via the main sidebar to

Player Management, Balance Configuration Management (View/Edit), Match Records, Bug & and Transaction Management.

[Placeholder: Insert Screens Flow Diagram for Unity Game Client and Web Admin Portal here]

3.1.2 Screen Descriptions

The table below describes all screens and interactive overlays across the Unity Game Client and the Web Admin Portal.

#	Feature	Screen	Description
1	Account & Profile Management	Login Screen	Entry screen of the Unity Client allowing Guests to sign in with username/password or automatically resume a valid saved JWT session.
2	Account & Profile Management	Register Screen	Form screen allowing Guests to register a new player account with username, email, and password validation.
3	Navigation & Account Management	Main Menu Screen	Central hub after authentication providing navigation to Create Room, Join Room by Code, Profile & Match History, Skin Store, Change Password, Bug Report Modal, and Logout.
4	Account & Profile Management	Profile & Match History Screen	Displays the authenticated Player's account details, aggregated survival/hunter statistics, Change Password dialog, and chronological match history.
5	Cosmetic Skin Store & Payment	Skin Store & Payment Screen	Displays available character cosmetic skins, ownership status, skin equip controls, and the payment checkout flow integrated with the Sandbox Payment Gateway.
6	Room & Lobby Management	Room Lobby Screen	Displays the 6-character room code, the list of up to 4 connected players, ready status indicators, Ready toggle button, Leave Room button, and Host-only Kick and Start Match controls.
7	Core Movement, Interaction & Map Exploration	In-Game HUD & Overlays	Displays role-appropriate gameplay HUD: Health, Sanity, Stamina, role ability cooldown/drone view, shared machine repair counter, interaction prompts, the

			on-screen Repair Minigame Overlay, and Bug Report / Reconnect prompts.
8	Match End & Results Processing	Match Result Screen	Post-match summary screen displaying Survivor Victory or Shade Victory, match duration, per-player role, final status (Escaped, Alive, Downed, Disconnected), and return to Main Menu button.
9	Web Admin Portal Management	Admin Login Screen	Web login screen authenticating administrator credentials and verifying ADMIN role privileges.
10	Web Admin Portal Management	Admin Dashboard Screen	Displays real-time platform KPIs, active rooms and dedicated servers, match completion counts, and Survivor vs. Shade win-rate charts.
11	Web Admin Portal Management	Player Management Screen	Allows Admins to search and filter registered players, inspect player profiles/stats, and activate or deactivate player accounts (ACTIVE / DEACTIVE).
12	Web Admin Portal Management	Balance Configuration Screen	Allows Admins to view versioned gameplay balance parameters (Sanity, Stamina, repair counts, Shade light debuffs) and publish updated configuration versions.
13	Web Admin Portal Management	Match Records & Bug Reports Screen	Allows Admins to inspect persisted match session records and review/resolve player-submitted bug reports.
14	Web Admin Portal Management	Transaction Management Screen	Allows Admins to search, filter, and reconcile cosmetic skin purchase transactions and payment gateway statuses.

3.1.3 Screen Authorization

The matrix below defines role-based access permissions across all system screens and major activities (X = Permitted).

Screen / System Activity	Guest	Player	Host	Survivor	Shade	Admin
Login Screen / Register Screen (UC-01, UC-02)	X					X

Main Menu / Logout / Change Password (UC-03, UC-04)		X	X			
Profile & Match History Screen (UC-14)		X	X			
Skin Store & Payment Screen (UC-15, UC-16)		X	X			
Create Room (UC-05)		X	X			
Join Room by Code (UC-06)		X				
Room Lobby: View Status / Set Ready / Leave (UC-07, UC-08, UC-11)		X	X			
Room Lobby: Kick Player / Start Match (UC-09, UC-10)			X			
In-Game HUD: Move, Open/Close Doors (UC-13)				X	X	
In-Game HUD: Repair Minigame, Sprint, Revive, Survivor Ability, Escape (UC-17..20)				X		
In-Game HUD: Hunt/Attack Survivors, Shade Ability (UC-21, UC-22)					X	
Match Result Screen (UC-23)		X	X	X	X	
Web Admin Portal: Dashboard, Players, Balance, Matches, Bug Reports, Transactions (UC-24..31)						X
Bug Report Modal (UC-32)		X	X	X	X	

3.1.4 Non-Screen Functions

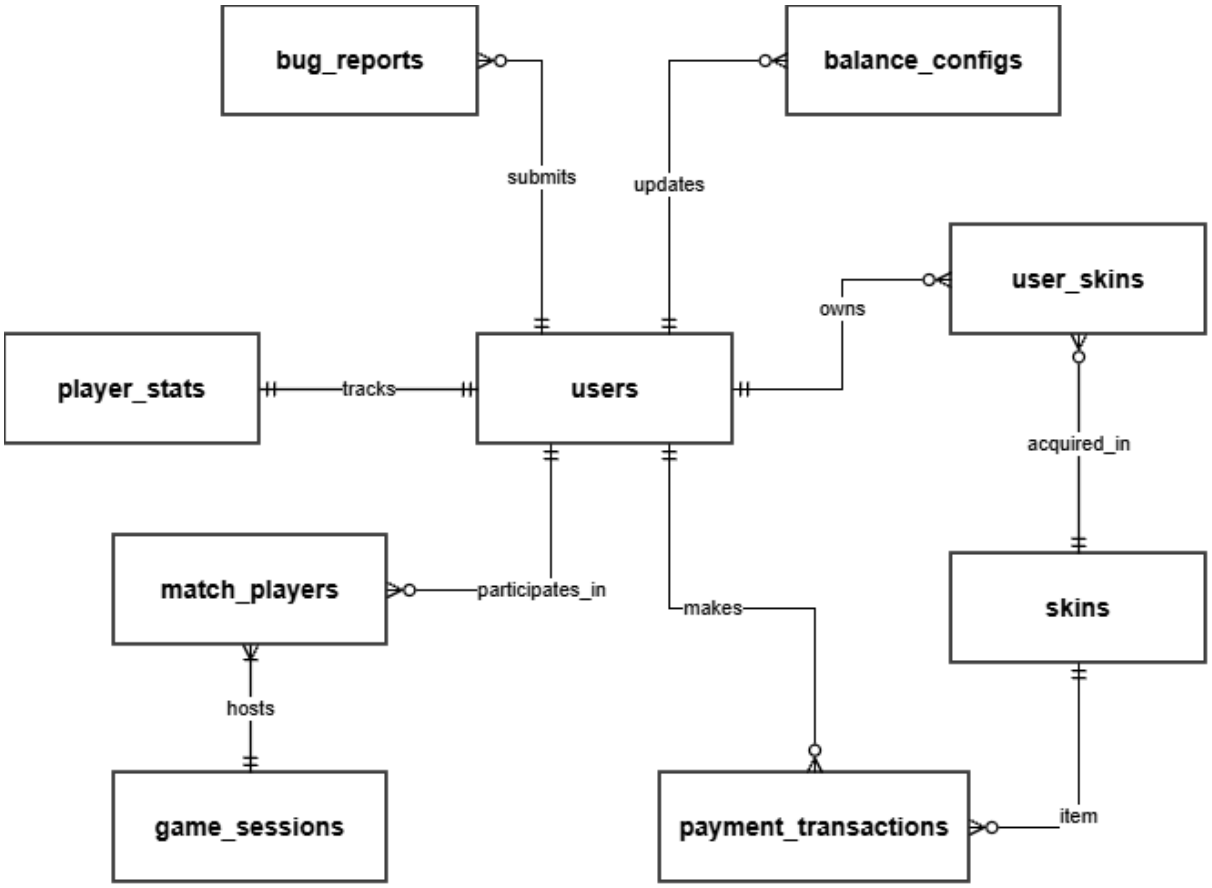
The table below describes the backend and authoritative dedicated server non-screen functions (REST APIs, background watchdogs, and server-side rule engines).

#	Feature	System Function	Description
1	Account & Profile Management	JWT Auth & Password Hashing Service	Validates credentials, hashes passwords and secrets with BCrypt, issues signed JWTs, and verifies bearer tokens for client and admin requests.

2	Room & Lobby Management	Room State & Host Migration Service	Manages room lifecycle (WAITING, STARTING, IN_GAME, CLOSED), enforces 4-player capacity, handles player leave/kick, and automatically migrates Host ownership if the Host leaves a WAITING room.
3	Match Session & Synchronization	Dedicated Server Orchestrator & Role Assigner	Allocates a headless Dedicated Game Server instance when the Host starts a match, randomly assigns 3 unique Survivor roles and 1 Shade role, and generates per-player join tokens.
4	Match Session & Synchronization	Join-Token Approval & Reconnect Service	Validates client connection payloads on the Dedicated Server against the Backend (/validate-token), admitting initial connections and allowing disconnected players to reconnect and reclaim their stationary character.
5	Match Session & Synchronization	Server Heartbeat & Empty-Server Watchdog	Periodically reports Dedicated Server health to the Backend (/heartbeat), enforces empty-server timeouts if all 4 players disconnect, and triggers the Reconnect Deadlock Safeguard when all connected Survivors have escaped or been downed.
6	Survivor Survival & Abilities	Server Gameplay Rules Engine	Executes server-authoritative calculations on the Dedicated Server for Sanity drain/debuffs, Shade light exposure debuffs, Health/Downed/Revive transitions, Stamina consumption, machine repair completion, and Emergency Exit unlocking.
7	Match End & Results Processing	Idempotent Match Result & Stats Service	Receives final match outcomes from the Dedicated Server (/result) using row-level locking and server-secret verification, persisting match records and updating aggregated PlayerStats in PostgreSQL.
8	Cosmetic Skin Store & Payment	Sandbox Payment Verification Service	Initiates payment sessions and processes payment gateway callbacks to atomically record transactions and unlock purchased skins in the player's inventory.

9	Bug & Issue Reporting	Bug Logging Service	Captures player-submitted bug/issue reports into PostgreSQL for Admin Portal review.

3.1.5 Entity Relationship Diagram



Entities Description:

#	Entity	Description
1	User	Stores registered account credentials, system role, account status and timestamps.
2	PlayerStats	Stores 1-to-1 aggregated career statistics for each User.
3	Room	Represents a 4-player pre-match room lobby identified by a unique 6-character room_code, tracking host_user_id, room status (WAITING, STARTING, IN_GAME, CLOSED), and member readiness.
4	GameSession	Represents an allocated match instance linked to a Dedicated Game Server.

5	MatchPlayer	Associates a User with a GameSession, storing assigned character_role, join_token_hash, connection/reconnect state, and final result status.
6	Skin	Catalog of purchasable character cosmetic skins.
7	UserSkin	Tracks skins unlocked and owned by each User, including whether the skin is currently equipped for that character role.
8	Transaction	Audit record of store purchase attempts and payment gateway callbacks.
9	BalanceConfig	Stores versioned gameplay balance parameters (repair target count, Sanity drain/regen rates, Stamina capacity, Shade light debuff multipliers) edited by Admins.
10	BugReport	Stores bug reports, gameplay issue reports submitted by players, reviewed by Admins.

3.2 Account & Profile Management

Provides player identity, credentials management, session establishment, password security, and career statistical tracking across the platform (UC-01, UC-02, UC-03, UC-04, UC-14).

3.2.1 Player Registration

Function Trigger: Triggered when an unauthenticated Guest clicks the Register button on the Login Screen.

Function Description:

- **Actors:** Guest
- **Purpose:** Enables new players to create an account with credentials verified by backend validation.
- **Interface:** Register Screen (Unity Client) communicating with POST /api/v1/auth/register.
- **Data Processing:** Validates uniqueness of username and email, computes BCrypt password hash, and inserts a new ACTIVE User record into PostgreSQL.

Screen Layout: [Placeholder: Insert Register Screen Mockup/Screenshot]

Function Details:

- **Data & Validation:** Username: 3 to 64 alphanumeric characters, required; Email: valid RFC 5322 format, max 255 characters, required; Password: 8 to 72 characters, required.
- **Business Rules:** BR-45 (Mandatory account authentication before accessing lobby or gameplay).
- **Normal Flow:** Guest fills in username, email, and password, then submits form; backend verifies data and creates account; client displays MSG-02 and navigates to Login Screen.

- **Abnormal Flow:** If username or email already exists, backend rejects with HTTP 409 and client displays MSG-21. If format validation fails, inline field validation error is displayed.

ID	Requirement	Priority
FR-AUTH-01	When launching the Unity Game Client, the system shall require the user to authenticate (Register or Login) before accessing the Main Menu, Room Lobby, Profile, or Store.	Must
FR-AUTH-02	A Guest shall be able to register a new account by providing a unique username, a valid unique email address, and a password meeting complexity rules.	Must

3.2.2 Player Login

Function Trigger: Triggered upon launching the Unity Client (automatic check) or when a Guest submits the login form.

Function Description:

- **Actors:** Guest, Player
- **Purpose:** Authenticates player identity, issues a signed JWT access token, and enables seamless subsequent access via stored credentials.
- **Interface:** Login Screen (Unity Client) communicating with POST /api/v1/auth/login and GET /api/v1/me.
- **Data Processing:** Verifies credentials against BCrypt hash, checks account status (enabled: ACTIVE vs. DEACTIVE), generates signed JJWT token, and loads user profile.

Screen Layout: [Placeholder: Insert Login & Auto-Login Screen Mockup/Screenshot]

Function Details:

- **Data & Validation:** Login identifier (username or email): required; Password: required; Saved JWT token: verified via signature and expiration timestamp.
- **Business Rules:** BR-45 (Authentication prerequisite and token persistence for automatic login).
- **Normal Flow:** Client checks for local stored token; if valid, validates via GET /api/v1/me and enters Main Menu directly (Auto-Login). Otherwise, Guest enters credentials and logs in; client stores token and transitions to Main Menu.
- **Abnormal Flow:** Incorrect credentials returns HTTP 401 with MSG-01. Expired token redirects to Login Screen with MSG-25. Deactivated account returns HTTP 403 with MSG-24.

ID	Requirement	Priority
FR-AUTH-03	A Guest shall be able to log in with a valid username/email and password to receive an authenticated JWT session.	Must

FR-AUTH-04	The Unity Game Client shall support automatic login on subsequent launches when a valid, non-expired authentication token is stored locally.	Must

3.2.3 Player Logout & Password Change

Function Trigger: Triggered when an authenticated Player selects Logout or Change Password from the Main Menu / Profile screen.

Function Description:

- **Actors:** Player
- **Purpose:** Allows players to safely terminate their local authenticated session or update their account password.
- **Interface:** Change Password Modal & Logout confirmation dialog on Unity Client.
- **Data Processing:** Clears local token cache on logout; verifies existing password before hashing and updating new password in database.

Screen Layout: [Placeholder: Insert Change Password Dialog Mockup/Screenshot]

Function Details:

- **Data & Validation:** Current Password: required; New Password: 8 to 72 characters, must differ from current password.
- **Business Rules:** BR-45 (Session credential lifecycle and account security).
- **Normal Flow:** Player enters current and new password; backend verifies existing password, updates hash, and returns success (MSG-22). Clicking Logout purges local token and returns to Login Screen.
- **Abnormal Flow:** Incorrect current password returns HTTP 400 with MSG-23. Password complexity failure displays inline error.

ID	Requirement	Priority
FR-AUTH-05	An authenticated Player shall be able to log out (clearing local session credentials) and shall be able to change their password after verifying their current password.	Must

3.2.4 Profile & Career Statistics

Function Trigger: Triggered when an authenticated Player selects Profile & Match History from the Main Menu.

Function Description:

- **Actors:** Player
- **Purpose:** Presents the player's aggregated career performance records and chronological match log.
- **Interface:** Profile & Match History Screen on Unity Client.
- **Data Processing:** Queries GET /api/v1/me and GET /api/v1/me/matches to retrieve PlayerStats and historical GameSession records.

Screen Layout: [Placeholder: Insert Profile & Match History Screen Mockup/Screenshot]

Function Details:

- **Data & Validation:** Read-only presentation of user profile, aggregated stats, and match history list.
- **Business Rules:** BR-43 (Persistent player stats and match records; no public leaderboard).
- **Normal Flow:** Player opens screen; client fetches and displays username, account creation date, career stats (matches, wins, escapes, revives, repairs), and scrollable match history.
- **Abnormal Flow:** Network timeout or authorization failure displays inline retry prompt.

ID	Requirement	Priority
FR-PRO-01	An authenticated Player shall be able to view their personal profile and aggregated career statistics (matches played, wins, escapes, repairs, revives) on the Unity Client.	Must
FR-PRO-02	An authenticated Player shall be able to view their chronological match history (including match date, assigned role, outcome, and duration) on the Unity Client without a global leaderboard.	Must

3.3 Room & Lobby Management

Enables authenticated players to assemble 4-player lobbies, share 6-character room codes, toggle readiness, and manage lobby state prior to match initiation (UC-05, UC-06, UC-07, UC-08, UC-09, UC-10, UC-11).

3.3.1 Create Room & Share Code

Function Trigger: Triggered when an authenticated Player clicks Create Room from the Main Menu.

Function Description:

- **Actors:** Host (Player)
- **Purpose:** Instantiates a new WAITING room on the backend, generates a unique 6-character room code, and establishes the creator as Host.
- **Interface:** Main Menu -> Room Lobby Screen via POST /api/rooms.
- **Data Processing:** Generates random 6-character code from unambiguous alphabet (A-Z, 0-9), initializes in-memory Room with maxPlayers=4, and sets creator as Host.

Screen Layout: [Placeholder: Insert Room Creation & Room Code Display Screenshot]

Function Details:

- **Data & Validation:** Room Code: exactly 6 characters matching ^[A-Z0-9]{6}$; Player capacity: 4.
- **Business Rules:** BR-05 (Unique 6-character room code), BR-06 (Max 4 players per room), BR-11 (Backend manages room state).

- **Normal Flow:** Player clicks Create Room; client displays MSG-03; backend confirms creation (MSG-04) and returns room code; client enters Room Lobby Screen.
- **Abnormal Flow:** If backend room service is unreachable, displays error modal with retry option.

ID	Requirement	Priority
FR-NET-01	The system shall support four authenticated players in an online multiplayer room.	Must
FR-NET-02	An authenticated player shall be able to create a server-hosted room as the Host and receive a unique shareable 6-character room code.	Must

3.3.2 Join Room by Code

Function Trigger: Triggered when a Player enters a 6-character code into the Join Room input and clicks Join.

Function Description:

- **Actors:** Player
- **Purpose:** Allows players to join an existing pre-match room lobby using a valid room code.
- **Interface:** Join Room Input Dialog on Main Menu via POST /api/rooms/{code}/join.
- **Data Processing:** Validates room code format, checks room status (must be WAITING) and player count (must be < 4), adds player to room member list.

Screen Layout: [Placeholder: Insert Join Room Code Input Modal Mockup/Screenshot]

Function Details:

- **Data & Validation:** Code input: 6 characters, case-insensitive, sanitized.
- **Business Rules:** BR-05, BR-06, BR-07 (One slot per player), BR-10 (No joins after match starts).
- **Normal Flow:** Player enters code and clicks Join; client displays MSG-05; backend admits player; client enters Room Lobby Screen.
- **Abnormal Flow:** Invalid, expired, in-game, or full room returns HTTP 400/404/409 with MSG-06.

ID	Requirement	Priority
FR-NET-03	An authenticated player shall be able to join an existing WAITING room by entering its valid room code, provided the room has fewer than four players.	Must

3.3.3 Lobby Readiness & Member Management

Function Trigger: Triggered when players in a room lobby toggle their ready state or when the lobby polling timer ticks (1.5s).

Function Description:

- **Actors:** Player, Host
- **Purpose:** Synchronizes member list, host identity, and ready statuses among all connected lobby participants.
- **Interface:** Room Lobby Screen on Unity Client via GET /api/rooms/{code} and POST /api/rooms/{code}/ready.
- **Data Processing:** Updates player readiness flag in Room state and returns current member snapshots to all polling clients.

Screen Layout: [Placeholder: Insert Room Lobby Member List & Ready Toggle Button Screenshot]

Function Details:

- **Data & Validation:** Ready state: boolean (true = Ready, false = Not Ready).
- **Business Rules:** BR-04, BR-08 (All players must be Ready before match start).
- **Normal Flow:** Player clicks Ready button; status toggles between Not Ready and Ready; all clients reflect updated checkmark.
- **Abnormal Flow:** Network drop during lobby triggers automatic polling reconnect.

ID	Requirement	Priority
FR-NET-04	The room lobby shall display the room code, all connected players, the Host indicator, and each player's ready state.	Must
FR-NET-10	Players in a WAITING room lobby shall be able to toggle their ready status (Ready / Not Ready).	Must

3.3.4 Kick Player

Function Trigger: Triggered when the Host clicks the Kick button adjacent to a non-host player in the Room Lobby.

Function Description:

- **Actors:** Host
- **Purpose:** Enables the room Host to remove an unwanted or inactive player from a WAITING room.
- **Interface:** Room Lobby Screen (Kick button visible to Host only) via POST /api/rooms/{code}/kick.
- **Data Processing:** Verifies caller is Host, removes target player from Room member list, and notifies kicked player on next poll.

Screen Layout: [Placeholder: Insert Host Kick Player Confirmation Popup Mockup/Screenshot]

Function Details:

- **Data & Validation:** Target playerId: must belong to current room and must not be the Host.
- **Business Rules:** BR-09 (Only Host may kick players in WAITING rooms).
- **Normal Flow:** Host clicks Kick on a player; target is removed from member list; target client receives MSG-09 and returns to Main Menu.

- **Abnormal Flow:** Non-host attempting kick receives HTTP 403 Forbidden.

ID	Requirement	Priority
FR-NET-12	The Host shall be able to kick a non-host player from the room lobby while the room is in the WAITING state.	Must

3.3.5 Leave Room & Host Migration

Function Trigger: Triggered when a Player or Host clicks Leave Room from the Room Lobby Screen.

Function Description:

- **Actors:** Player, Host
- **Purpose:** Handles voluntary exit from a lobby with automatic Host migration if the departing player is Host.
- **Interface:** Room Lobby Screen via POST /api/rooms/{code}/leave.
- **Data Processing:** Removes player from Room. If departing player was Host and room is not empty, assigns Host role to next remaining player; if room empty, closes and purges Room.

Screen Layout: [Placeholder: Insert Leave Room & Host Migration Notification Screenshot]

Function Details:

- **Data & Validation:** PlayerId: valid member of room.
- **Business Rules:** BR-46 (Automatic host migration upon host departure; close room if empty).
- **Normal Flow:** Player clicks Leave; client returns to Main Menu. If Host leaves, remaining members receive MSG-08 and host badge updates to new host.
- **Abnormal Flow:** Leaving an already closed room redirects cleanly to Main Menu.

ID	Requirement	Priority
FR-NET-13	A player shall be able to leave a WAITING room before match start; if the Host leaves, the system shall automatically transfer Host authority to the next remaining player in the room, or close the room if no players remain.	Must

3.3.6 Initiate Match Start

Function Trigger: Triggered when the Host clicks Start Match button in a fully prepared Room Lobby.

Function Description:

- **Actors:** Host
- **Purpose:** Validates start prerequisites and transitions the room state from WAITING to STARTING, initiating dedicated server allocation.
- **Interface:** Room Lobby Screen via POST /api/rooms/{code}/start.
- **Data Processing:** Enforces 4-player count and all-ready conditions; locks room against joins; triggers match orchestration.

Screen Layout: [Placeholder: Insert Start Match Button & Allocating Server Banner Mockup/Screenshot]

Function Details:

- **Data & Validation:** Room capacity: exactly 4 players; Readiness: 4/4 players marked Ready.
- **Business Rules:** BR-08 (Prerequisites for match initiation), BR-09 (Only Host may start), BR-10 (Room locked upon start).
- **Normal Flow:** Host clicks Start Match; client displays MSG-10; room transitions to STARTING; orchestrator allocates Dedicated Server.
- **Abnormal Flow:** If fewer than 4 players or any player is Not Ready, start is rejected and client displays MSG-27.

ID	Requirement	Priority
FR-NET-11	The Host shall be able to start the match only when the room has exactly four connected players and all four players are marked Ready.	Must

3.4 Match Session & Synchronization

Coordinates dedicated game server allocation, 3v1 role assignment, client connection approval, server-authoritative state synchronization, disconnect retention, and session reliability (UC-09, UC-12).

3.4.1 Random Role Assignment

Function Trigger: Triggered automatically by the backend MatchService when the Host initiates match start.

Function Description:

- **Actors:** Survivor, Shade Player (System-automated)
- **Purpose:** Randomly assigns exactly one player to The Shade and the remaining three players to unique Survivor roles (Drone Operator, Engineer, Logistic).
- **Interface:** N/A - non-screen function (orchestrated by Backend MatchService).
- **Data Processing:** Shuffles player IDs, randomly selects 1 Shade, distributes 3 unique Survivor roles without duplicates, stores assignments in MatchPlayer records.

Screen Layout: N/A - non-screen function

Function Details:

- **Data & Validation:** Role set: exactly {Shade, DroneOperator, Engineer, Logistic}; Player count: exactly 4.
- **Business Rules:** BR-01 (3 Survivors + 1 Shade), BR-02 (Random assignment), BR-03 (Unique Survivor roles), BR-04 (Role immutable for match duration).
- **Normal Flow:** System executes shuffle and assignment; returns assigned role and joinToken to each client; clients display MSG-11 upon match load.
- **Abnormal Flow:** If player count is not exactly 4, match allocation is aborted.

ID	Requirement	Priority

FR-NET-05	When the match begins, the system shall randomly assign three players to Survivor roles and one player to The Shade.	Must
FR-NET-06	Role assignment shall be unique, with each of the three Survivor roles (Drone Operator, Engineer, Logistic) assigned to at most one player per match.	Must
FR-NET-07	Players shall not manually select or change their assigned role before or after role assignment.	Must

3.4.2 Dedicated Server Allocation & Join-Token Approval

Function Trigger: Triggered when the Dedicated Server container boots and clients connect via Unity Transport (UTP).

Function Description:

- **Actors:** Player (System-automated)
- **Purpose:** Allocates headless Dedicated Game Server instance, generates cryptographic server secret and single-use join tokens, and validates connections.
- **Interface:** N/A - non-screen function (Backend orchestrator & Dedicated Server ServerConnectionApproval).
- **Data Processing:** Backend stores GameSession with hashed server secret; Dedicated Server validates connection payload via POST /api/v1/matches/{id}/validate-token.

Screen Layout: N/A - non-screen function

Function Details:

- **Data & Validation:** Join Token: cryptographically random, verified against BCrypt hash; GameSession status: RUNNING.
- **Business Rules:** BR-11 (Dedicated server authority), BR-36, BR-37, BR-38 (Connection credentials issued only after server ready).
- **Normal Flow:** Clients connect using IP, port, and joinToken; Dedicated Server verifies token with backend; connection approved; player spawned.
- **Abnormal Flow:** Invalid or expired joinToken causes connection rejection with DisconnectReason: InvalidToken.

ID	Requirement	Priority
FR-NET-09	The authoritative dedicated game server shall validate role assignment, player state, machine state, exit state, Health, Sanity, Stamina, and match outcome.	Must
FR-NET-14	The backend shall allocate a dedicated game server instance upon match start and issue per-player join tokens; the dedicated server shall verify each client's join token with the backend before approving the network connection or reconnection.	Must

3.4.3 Disconnect, Reconnect & Character Retention

Function Trigger: Triggered when a player client experiences network drop or reconnects during an active match.

Function Description:

- **Actors:** Survivor, Shade Player
- **Purpose:** Retains disconnected player's character stationary in-game to preserve match continuity and supports reconnecting to resume character control.
- **Interface:** In-Game Reconnect Overlay on Unity Client (MSG-19).
- **Data Processing:** Dedicated Server detects disconnect, keeps NetworkObject alive, zeros velocity, cancels active minigames; upon reconnection with valid credentials, calls ChangeOwnership to new clientId.

Screen Layout: [Placeholder: Insert Reconnecting to Match Overlay Screenshot]

Function Details:

- **Data & Validation:** Match state: RUNNING; Reconnect window: subject to configured reconnect timeout.
- **Business Rules:** BR-47 (Stationary character retention, reconnect authorization, and deadlock safeguard).
- **Normal Flow:** Player disconnects; character stops and stands still; teammates can protect/revive them; Shade can hunt them. Player relaunches, client sends reconnect request, server reclaims character.
- **Abnormal Flow:** If all remaining connected Survivors have escaped or been downed, server applies Deadlock Safeguard and finalizes match without waiting for disconnected player.

ID	Requirement	Priority
FR-NET-08	When a Survivor or Shade player loses connection during an active match, their character shall remain stationary in-game and take no action while the match continues for the remaining players; the system shall allow the disconnected player to reconnect and resume control of their character, and shall automatically conclude the match if all remaining connected Survivors have either Escaped or been Downed (or upon reconnect timeout).	Must

3.4.4 Server Heartbeat & Session Watchdog

Function Trigger: Triggered periodically (every 5 seconds) on Dedicated Server tick loop.

Function Description:

- **Actors:** Hosting Infrastructure (System-automated)
- **Purpose:** Monitors dedicated server health, reports active match telemetry to backend, and enforces empty-server timeouts.
- **Interface:** N/A - non-screen function (Dedicated Server loop via POST /api/v1/matches/{id}/heartbeat).

- **Data Processing:** Sends connected player count, tick rate, and match uptime; updates lastHeartbeatAt; aborts session if empty for longer than timeout.

Screen Layout: N/A - non-screen function

Function Details:

- **Data & Validation:** Heartbeat interval: 5s; Empty server timeout: default 60s.
- **Business Rules:** BR-40 (Handling connection failures and aborting abandoned sessions).
- **Normal Flow:** Dedicated Server sends heartbeat; backend updates session health; match proceeds normally.
- **Abnormal Flow:** If all 4 players disconnect and remain absent beyond timeout, server aborts session and terminates container.

ID	Requirement	Priority
FR-NET-15	The dedicated game server shall send periodic health heartbeats to the backend during an active match and automatically abort the session if all four players remain disconnected beyond the configured empty-server timeout.	Should

3.5 Core Movement, Interaction & Map

Governs 2D top-down locomotion, physical collisions, interactive door mechanics, map spatial structure, and spawn zones (UC-13).

3.5.1 Character Movement & Camera Control

Function Trigger: Triggered continuously during gameplay input loop.

Function Description:

- **Actors:** Survivor, Shade Player
- **Purpose:** Provides responsive 8-directional movement with physics collision and smooth camera tracking.
- **Interface:** In-Game Viewport on Unity Client.
- **Data Processing:** Reads keyboard inputs (WASD / Arrows), applies velocity via Rigidbody2D, updates facing direction and animation state, clamps Cinemachine camera to map bounds.

Screen Layout: [Placeholder: Insert In-Game Top-Down Movement & Camera Viewport Screenshot]

Function Details:

- **Data & Validation:** Input axes: horizontal [-1, 1], vertical [-1, 1]; Base speed: configured per character role.
- **Business Rules:** BR-12, BR-26 (Valid attribute and movement parameters).
- **Normal Flow:** Player presses movement keys; character moves in 8 directions with walking animations; camera tracks smoothly without leaving map bounds.
- **Abnormal Flow:** Colliding with obstacles or closed doors stops movement without clipping.

ID	Requirement	Priority
FR-CORE-01	Players shall move in eight directions using the keyboard.	Must
FR-CORE-02	The player shall collide with walls, closed doors, and obstacles.	Must
FR-CORE-03	The player character shall face its movement direction and show distinct idle and moving animations.	Must
FR-CORE-06	The camera shall follow the local player smoothly and stay inside the map boundaries.	Must

3.5.2 Object Interaction & Door Control

Function Trigger: Triggered when a player approaches an interactive door or object and presses Interact (E key).

Function Description:

- **Actors:** Survivor, Shade Player
- **Purpose:** Detects nearby interactable objects, displays prompt, and toggles door open/closed state on authoritative server.
- **Interface:** In-Game HUD Interaction Prompt (E key prompt).
- **Data Processing:** Client detects collider overlap, displays prompt; pressing E sends ServerRpc; Dedicated Server verifies distance and toggles door state replicated via NetworkVariable.

Screen Layout: [Placeholder: Insert Interactive Door & Interaction Prompt HUD Screenshot]

Function Details:

- **Data & Validation:** Interaction distance: maximum 2.0 Unity units.
- **Business Rules:** BR-36 (Authoritative interaction validation).
- **Normal Flow:** Player approaches closed door; prompt appears; player presses E; door opens for all players; player can press E again to close door.
- **Abnormal Flow:** Attempting interaction while out of range or Downed is rejected by server.

ID	Requirement	Priority
FR-CORE-04	The system shall detect interactable objects near the player and show an interaction prompt.	Must
FR-CORE-05	Interactions shall be validated by the authoritative server before taking effect.	Must
FR-MAP-10	Rooms and corridors shall include interactive doors that both Survivors and The Shade can open and close; The Shade shall not be physically blocked from entering the Safe Zone by an invisible barrier, but shall suffer the configured light debuff while inside its illuminated area.	Must

3.5.3 Station Map Layout & Spawn Configuration

Function Trigger: Triggered upon loading the gameplay scene at match start.

Function Description:

- **Actors:** Survivor, Shade Player (System-automated)
- **Purpose:** Initializes station geometry, spawns Survivors inside Safe Zone, spawns The Shade at dedicated distant point, and locks Emergency Exits.
- **Interface:** N/A - non-screen function (Scene load & spawn orchestration).
- **Data Processing:** Positions 3 Survivors at Safe Zone spawn points; positions Shade at Hunter spawn point; instantiates doors, broken machines, lighting zones, and locked exits.

Screen Layout: [Placeholder: Insert Space Station Full Map Layout Overview Screenshot]

Function Details:

- **Data & Validation:** Spawn coordinates: configured spawn markers; Emergency Exit rooms: exactly 2.
- **Business Rules:** BR-27 (Predefined broken machines), BR-28 (Locked exits at start), BR-29 (Safe Zone spawn).
- **Normal Flow:** Scene loads; all players spawn at correct starting locations; Safe Zone is lit; exits are locked.
- **Abnormal Flow:** If spawn points are obstructed, server uses fallback offset coordinates.

ID	Requirement	Priority
FR-MAP-01	The game shall provide one multi-room space station map with rooms connected by corridors and alternate loop pathways.	Must
FR-MAP-04	The map shall contain one illuminated Survivor Safe Zone where all three Survivors spawn at match start.	Must
FR-MAP-05	The map shall contain two Emergency Exit rooms located at distinct extraction points.	Must
FR-MAP-09	The map shall provide a dedicated spawn point for The Shade located at a safe distance outside the Survivor Safe Zone.	Must
FR-OBJ-01	At match start, four players shall enter the map: three Survivors spawning in the Safe Zone and one Shade spawning at the Hunter spawn point.	Must

3.6 Survivor Survival & Role Abilities

Manages Survivor health, the Downed crawl state, teammate revive interactions, stamina sprinting, and distinct abilities for Drone Operator, Engineer, and Logistic (UC-18, UC-19, UC-20).

3.6.1 Health, Downed State & Teammate Revive

Function Trigger: Triggered when a Survivor takes damage reducing Health to 0, or when a teammate channels Revive on a Downed player.

Function Description:

- **Actors:** Survivor
- **Purpose:** Transitions Survivor to Downed state upon zero health and allows living teammates to restore them via channeled revive.
- **Interface:** Survivor HUD (Health Bar, Downed Indicator, Revive Progress Bar).
- **Data Processing:** Server monitors Health; at 0 HP, sets state to Downed, reduces movement speed to 35%, disables abilities/repair; during revive channel, accumulates revive progress until restoring 35% HP.

Screen Layout: [Placeholder: Insert Survivor Health Bar, Downed Crawling & Revive HUD Screenshot]

Function Details:

- **Data & Validation:** Health: 0 to 100; Downed crawl speed: 35% of base; Revive duration: default 3.0 seconds; Revived HP: 35% of maximum.
- **Business Rules:** BR-12, BR-13 (Downed state), BR-48.
- **Normal Flow:** Survivor takes lethal damage and enters Downed state; living teammate approaches and holds E; revive channel completes; player stands up with 35% HP.
- **Abnormal Flow:** Taking damage while reviving interrupts the channel. If all 3 Survivors are simultaneously Downed, match ends immediately.

ID	Requirement	Priority
FR-SUR-01	Each Survivor shall have a Health value and enter the Downed state when their Health reaches zero; a Downed Survivor shall move at a reduced crawl speed (default 35% of base speed), shall not bleed out over time, and shall be unable to repair machines or trigger extraction alone.	Must
FR-SUR-02	Living teammates shall be able to revive a Downed Survivor through a server-validated channeled interaction, restoring the Survivor to the Alive state with partial Health (default 35% of maximum Health).	Must

3.6.2 Stamina & Sprinting

Function Trigger: Triggered when a living Survivor holds Left Shift while moving.

Function Description:

- **Actors:** Survivor
- **Purpose:** Provides a temporary movement speed boost consuming Stamina, with passive regeneration when not sprinting.
- **Interface:** Survivor HUD (Stamina Bar).
- **Data Processing:**

Screen Layout: [Placeholder: Insert Survivor Stamina Bar & Sprint Visual Indicator Screenshot]

Function Details:

- **Data & Validation:** Stamina: 0 to 100 (higher for Engineer)
- **Business Rules:** BR-14 (Stamina consumption during sprint), BR-16 (Engineer higher stamina).
- **Normal Flow:** Player holds Shift; character sprints faster; Stamina bar depletes; releasing Shift allows Stamina to refill.
- **Abnormal Flow:** Depleting Stamina to 0 drops character to normal walking speed until at least 20% Stamina regenerates.

ID	Requirement	Priority
FR-SUR-03	Each Survivor shall have a Stamina value that limits sustained sprinting and regenerates when the Survivor is not sprinting.	Must
FR-SUR-04	Living Survivors shall be able to sprint, consuming Stamina while sprinting.	Must

3.6.3 Role Specialization & Abilities (Drone Operator, Engineer, Logistic)

Function Trigger: Triggered when a Survivor presses the Role Ability key (Q or F) or passively during machine repair.

Function Description:

- **Actors:** Survivor (Drone Operator, Engineer, Logistic)
- **Purpose:** Executes unique role mechanics: Drone Operator deploys recon drone, Engineer repairs faster with higher stamina, Logistic stuns The Shade.
- **Interface:** Survivor HUD (Ability Icon, Cooldown Timer, Drone Recon Overlay).
- **Data Processing:** Drone Operator switches camera to steerable drone with spotlight/radar; Engineer applies repair multiplier; Logistic triggers area stun RPC applying 3s immobilized debuff to Shade.

Screen Layout: [Placeholder: Insert Drone Recon Overlay & Logistic Stun VFX Screenshot]

Function Details:

- **Data & Validation:** Drone duration/battery: 30s; Logistic stun duration: 3.0s.
- **Business Rules:** BR-15, BR-16, BR-17, BR-18, BR-19 (Strict role ability enforcement).
- **Normal Flow:** Player activates ability; server validates role and cooldown; effect executes; cooldown timer begins.
- **Abnormal Flow:** Activating ability on cooldown or while Downed is rejected.

ID	Requirement	Priority
FR-SUR-05	Each Survivor role (Drone Operator, Engineer, Logistic) shall provide its assigned player with the corresponding role-specific ability and attributes.	Must

FR-SUR-06	The Drone Operator shall pilot a recon drone that provides a steerable directional spotlight and a motion radar.	Must
FR-SUR-07	The Engineer shall repair broken machines faster than other Survivor roles under equivalent conditions.	Must
FR-SUR-08	The Logistic shall have a role-specific ability that can temporarily stun The Shade when activated within valid range and cooldown rules.	Must
FR-SUR-10	The Engineer shall have a higher maximum Stamina capacity than the other Survivor roles.	Must

3.7 The Shade - Hunter Gameplay

Governs the playable Hunter role, melee attacks, Hunter tracking abilities, and light-exposure debuffs (UC-21, UC-22).

3.7.1 Hunting & Downing Survivors

Function Trigger: Triggered when the Shade Player presses Primary Attack (Left Click) near a Survivor.

Function Description:

- **Actors:** The Shade (Shade Player)
- **Purpose:** Executes melee attacks dealing damage to Survivors with the objective of downing all three.
- **Interface:** The Shade HUD (Attack Crosshair, Target Feedback).
- **Data Processing:** Dedicated Server validates attack hitbox, applies damage to target Survivor, triggers attack cooldown animation.

Screen Layout: [Placeholder: Insert The Shade First-Person/Top-Down Attack View Screenshot]

Function Details:

- **Data & Validation:**
- **Business Rules:** BR-01, BR-33 (The Shade wins when all Survivors are Downed).
- **Normal Flow:** Shade pursues Survivor and strikes; Survivor loses HP; second hit downs Survivor.
- **Abnormal Flow:** Attacking while stunned by Logistic is blocked by server.

ID	Requirement	Priority
FR-HUN-01	The Shade shall be the Hunter role controlled by one player during a match.	Must
FR-HUN-02	The Shade shall be able to attack living Survivors to reduce their Health and attempt to down all of them.	Must
FR-HUN-04	The current game scope shall contain one playable Hunter type, The Shade.	Must

3.7.2 Hunter Abilities & Server Validation

Function Trigger: Triggered when the Shade Player activates Hunter Skill (Right Click or Space).

Function Description:

- **Actors:** The Shade (Shade Player)
- **Purpose:** Executes Hunter-specific tracking or mobility skill to locate and intercept Survivors.
- **Interface:** The Shade HUD (Skill Icon & Cooldown Indicator).
- **Data Processing:** Dedicated Server validates activation, applies movement buff or sensory highlights, synchronizes cooldown.

Screen Layout: [Placeholder: Insert The Shade Ability Activation & Cooldown HUD Screenshot]

Function Details:

- **Data & Validation:**
- **Business Rules:** BR-36 (Authoritative server validation of Hunter actions).
- **Normal Flow:** Shade activates skill; receives temporary sensory cue or speed surge; pursues targets.
- **Abnormal Flow:** Activation on cooldown is rejected by server.

ID	Requirement	Priority
FR-HUN-03	The Shade shall have its own movement profile and Hunter-specific abilities.	Must
FR-HUN-06	The authoritative dedicated server shall validate all of The Shade's attacks, ability activations, stun states, and light-debuff states.	Must

3.7.3 Light Exposure Debuff

Function Trigger: Triggered whenever The Shade steps into an illuminated area (Safe Zone, lit room, repaired machine room).

Function Description:

- **Actors:** The Shade (System-automated)
- **Purpose:** Applies movement speed reduction and vision impairment debuffs to The Shade while exposed to light.
- **Interface:** The Shade HUD (Light Burn/Debuff Screen Border Effect).
- **Data Processing:** Dedicated Server checks Shade collider against 2D light trigger zones; while overlapping, applies configured debuff multiplier to Shade speed.

Screen Layout: [Placeholder: Insert The Shade Light Debuff Vignette & Slow Indicator Screenshot]

Function Details:

- **Data & Validation:** Debuff multiplier: default 0.75x speed; Debuff clear delay: 0.5s after exiting light.
- **Business Rules:** BR-25 (Light exposure debuff to The Shade).
- **Normal Flow:** Shade enters illuminated room; screen border burns red; movement speed decreases; exiting room restores normal speed.
- **Abnormal Flow:** N/A (continuous environmental effect).

ID	Requirement	Priority
FR-HUN-05	The Shade shall suffer a configurable debuff when exposed to light generated by repaired machine rooms, the Safe Zone, and other illuminated areas.	Must

3.8 Machine Repair & Exit Objectives

Covers the core cooperative objective: completing repair minigames on broken machines, room illumination, and unlocking Emergency Exits (UC-17).

3.8.1 Broken Machine Repair Minigame

Function Trigger: Triggered when a living Survivor approaches an unrepaired broken machine and presses Interact (E key).

Function Description:

- **Actors:** Survivor
- **Purpose:** Opens an on-screen interactive repair minigame; successful completion marks machine repaired and restores room lighting.
- **Interface:** On-Screen Repair Minigame Overlay on Unity Client.
- **Data Processing:** Client renders minigame; Survivor completes skill checks; client sends ServerRpc; server validates completion, marks machine repaired, illuminates room, increments repair count.

Screen Layout: [Placeholder: Insert On-Screen Machine Repair Minigame UI Overlay Screenshot]

Function Details:

- **Data & Validation:** Minigame progress: 0 to 100%; Engineer completes at accelerated rate; Machine states: BROKEN, REPAIRED.
- **Business Rules:** BR-20, BR-21 (Server confirms repair), BR-22 (Repaired room illuminated), BR-51 (Failed/cancelled minigame can be retried).
- **Normal Flow:** Survivor interacts with machine; completes minigame; machine sparks clear and room lights turn on (MSG-12); shared repair counter increments.
- **Abnormal Flow:** Failing minigame or releasing interact cancels repair, triggers warning (MSG-13), alerts Shade, and leaves machine broken for retry.

ID	Requirement	Priority
FR-PUZ-01	The map shall contain a predefined number of broken machines at match start.	Must

FR-PUZ-02	Any living Survivor may interact with and repair a broken machine via an on-screen repair minigame regardless of their assigned role.	Must
FR-PUZ-03	A machine shall be considered repaired only after its on-screen repair minigame is successfully completed and validated by the server.	Must
FR-PUZ-04	When a Survivor fails or cancels an active repair minigame, the machine repair shall not be completed, the system shall apply the configured failure penalty or alert, and the machine shall remain available for subsequent repair attempts.	Must
FR-PUZ-05	The authoritative dedicated server shall validate machine repair completion and prevent duplicate completion of an already repaired machine.	Must
FR-MAP-02	[Merged into FR-PUZ-01] The map shall contain a predefined number of broken machines that can be repaired by any Survivor.	Must
FR-SUR-09	[Merged into FR-PUZ-02] All living Survivors shall be able to interact with and repair broken machines regardless of their assigned role.	Must

3.8.2 Shared Repair Progress & Emergency Exit Unlock

Function Trigger: Triggered when a machine repair is validated by the server.

Function Description:

- **Actors:** Survivor (System-automated)
- **Purpose:** Replicates repaired machine count to all HUDs; unlocks both Emergency Exit rooms when repair count reaches n.
- **Interface:** Survivor HUD Objective Tracker & Exit Door Status.
- **Data Processing:** Dedicated Server increments NetworkVariable repairedCount; checks against requiredCount (n); when reached, opens both Emergency Exit colliders and broadcasts MSG-14.

Screen Layout: [Placeholder: Insert Objective Counter (n/n Repaired) & Unlocked Exit Door Screenshot]

Function Details:

- **Data & Validation:** Required repair count n: configured in BalanceConfig (default 3 or 4); Total machines on map: >= n.
- **Business Rules:** BR-28 (Emergency exits remain locked until n machines repaired).
- **Normal Flow:** Survivors repair machines until reaching n; siren sounds; HUD announces exits unlocked (MSG-14); Emergency Exit doors become accessible.
- **Abnormal Flow:** Interacting with exit door before n repairs displays locked message.

ID	Requirement	Priority
FR-PUZ-06	The number of repaired machines shall be tracked and replicated to all players as shared match progress.	Must
FR-PUZ-07	When the required number of machines (n) is repaired, both Emergency Exit rooms shall unlock and become accessible.	Must
FR-OBJ-02	Survivors shall cooperate to repair the required number of broken machines (n).	Must
FR-MAP-07	[Merged into FR-PUZ-07 / FR-OBJ-03] Both Emergency Exit rooms shall remain locked at match start and become accessible when the required number of machines has been repaired.	Must
FR-OBJ-03	[Merged into FR-PUZ-07] Reaching the required repair count (n) shall unlock both Emergency Exit rooms.	Must

3.9 Darkness, Lighting & Sanity

Controls environmental darkness, 2D dynamic lighting zones, Survivor sanity depletion, and low-sanity psychological debuffs (UC-13, UC-17).

3.9.1 Lighting Zones & Room Illumination

Function Trigger: Triggered continuously based on player position and machine repair status.

Function Description:

- **Actors:** Survivor, The Shade (System-automated)
- **Purpose:** Maintains dark vs. illuminated zone boundaries; illuminated zones halt sanity loss and apply Shade debuffs.
- **Interface:** In-Game 2D Lighting Environment (Universal Render Pipeline 2D Lights).
- **Data Processing:** 2D Light components illuminate Safe Zone, permanently lit areas, and repaired rooms; triggers update player light exposure status.

Screen Layout: [Placeholder: Insert Comparison of Dark Corridor vs. Illuminated Repaired Room Screenshot]

Function Details:

- **Data & Validation:** Light types: Permanent Safe Zone light, Corridor ambient, Repaired Machine room light.
- **Business Rules:** BR-22, BR-25, BR-29, BR-30 (Lighting distribution and Shade debuff rules).
- **Normal Flow:** Survivor enters illuminated room; darkness vignette recedes; Sanity drain stops. Shade enters; light debuff applies.
- **Abnormal Flow:** Power outages or dynamic light flicker may be configured in BalanceConfig.

ID	Requirement	Priority

FR-MAP-03	The map shall contain multiple dark areas that affect Survivor Sanity.	Must
FR-MAP-06	The map shall contain several permanently lit areas in addition to the Safe Zone.	Must
FR-MAP-08	A repaired machine shall cause its room to become illuminated and shall apply a configurable debuff to The Shade while The Shade is exposed to that room's light.	Must
FR-SAN-01	Environmental darkness shall affect Survivors only.	Must

3.9.2 Sanity Drain, Thresholds & Debuffs

Function Trigger: Triggered every tick on Dedicated Server for each active Survivor.

Function Description:

- **Actors:** Survivor
- **Purpose:** Depletes Survivor Sanity while in darkness; triggers auditory/visual hallucinations and gameplay debuffs at low levels.
- **Interface:** Survivor HUD (Sanity Meter, Post-Processing Vignette/Distortion).
- **Data Processing:** Decreases Sanity at drainRate while in darkness; recovers Sanity in light; applies screen-shake and distortion below threshold.

Screen Layout: [Placeholder: Insert Low Sanity Visual Distortion & Audio Debuff Feedback Screenshot]

Function Details:

- **Data & Validation:** Sanity: 0 to 100; Drain rate: default 2%/s; Low Sanity threshold: default <= 30%.
- **Business Rules:** BR-23, BR-24, BR-26 (Sanity depletion in darkness and low sanity debuffs; Shade has no Sanity).
- **Normal Flow:** Survivor wanders in dark corridors; Sanity depletes; breathing intensifies; below 30%, screen distorts and auditory hallucinations occur; returning to light stabilizes Sanity.
- **Abnormal Flow:** Reaching 0 Sanity applies severe movement debuff or involuntary auditory screams.

ID	Requirement	Priority
FR-SAN-02	A Survivor's Sanity shall decrease while staying in dark areas.	Must
FR-SAN-03	The Shade shall not have a Sanity value and shall not suffer darkness penalties.	Should
FR-SAN-04	When a Survivor's Sanity falls below a configured threshold, one or more configured gameplay debuffs shall be applied.	Must
FR-SAN-05	Sanity shall remain clamped within its configured valid range (0 to maximum Sanity).	Could

FR-SAN-06	Illuminated areas (the Safe Zone, permanently lit rooms, and repaired machine rooms) shall prevent or reduce Survivor Sanity loss according to the configured balance rules.	Should

3.10 Match End & Results

Governs individual Survivor extraction, win condition evaluation, authoritative result reporting to PostgreSQL, and post-match result presentation (UC-13, UC-23).

3.10.1 Individual Survivor Extraction

Function Trigger: Triggered when a non-downed Survivor enters an unlocked Emergency Exit trigger zone.

Function Description:

- **Actors:** Survivor
- **Purpose:** Transitions the extracting Survivor to Escaped state, removes their character from active map, while remaining active players continue match.
- **Interface:** Emergency Exit Zone on Unity Client (MSG-16).
- **Data Processing:** Dedicated Server validates exit is unlocked and player is not Downed; updates player status to Escaped; disables character collider; keeps client in spectator view.

Screen Layout: [Placeholder: Insert Emergency Exit Extraction Zone & Survivor Escaped Notification Screenshot]

Function Details:

- **Data & Validation:** Exit status: UNLOCKED; Survivor status: ALIVE (not Downed).
- **Business Rules:** BR-31 (Individual escape transition), BR-32 (Match continues until all active survivors escape or downed).
- **Normal Flow:** Survivor reaches unlocked Emergency Exit; steps inside; receives MSG-16; transitions to Escaped; remaining teammates continue playing.
- **Abnormal Flow:** Downed Survivor crawling into exit cannot extract alone.

ID	Requirement	Priority
FR-OBJ-04	When an Emergency Exit is unlocked, any non-downed Survivor who enters the exit zone shall transition to the Escaped state and be removed from active map play while remaining active Survivors continue playing; the Survivor side shall win the match if at least one Survivor successfully escapes.	Must

3.10.2 Win Condition & Match Termination

Function Trigger: Triggered on Dedicated Server tick loop when all Survivors reach terminal states.

Function Description:

- **Actors:** Survivor, The Shade (System-automated)

- **Purpose:** Evaluates win conditions without a gameplay time limit: Survivor Victory if >= 1 escaped; Shade Victory if all 3 downed with 0 escapes.
- **Interface:** N/A - non-screen function (Dedicated Server MatchLifecycle).
- **Data Processing:** Checks terminal states: if all 3 Survivors are Downed (0 escaped), declares ShadeVictory; if all connected Survivors are either Escaped or Downed (with >= 1 Escaped), declares SurvivorVictory.

Screen Layout: N/A - non-screen function

Function Details:

- **Data & Validation:** Outcome: SurvivorVictory or ShadeVictory; Time limit: none.
- **Business Rules:** BR-32, BR-33, BR-34, BR-35 (Win conditions, no time limit, idempotent single termination).
- **Normal Flow:** Condition met; server freezes simulation; broadcasts result to all clients; initiates result reporting.
- **Abnormal Flow:** If a disconnected Survivor remains stationary past grace timeout, server evaluates remaining active players.

ID	Requirement	Priority
FR-OBJ-05	The Shade shall win the match when all three Survivors are Downed before any Survivor escapes.	Must
FR-OBJ-06	The match shall not use a gameplay time limit.	Must
FR-OBJ-07	The match shall end immediately and idempotently when all three Survivors are Downed (Shade Victory) or when all connected Survivors have either Escaped or been Downed with at least one Survivor Escaped (Survivor Victory).	Must

3.10.3 Authoritative Result Reporting & Persistence

Function Trigger: Triggered immediately upon match termination by Dedicated Server.

Function Description:

- **Actors:** Hosting Infrastructure (System-automated)
- **Purpose:** Reports final outcome, duration, and per-player metrics to Backend REST API for persistent storage in PostgreSQL.
- **Interface:** N/A - non-screen function (POST /api/v1/matches/{id}/result with X-Server-Secret).
- **Data Processing:** Dedicated Server sends payload; Backend validates secret, row-locks GameSession, updates status to COMPLETED, records MatchPlayer outcomes, updates aggregated PlayerStats.

Screen Layout: N/A - non-screen function

Function Details:

- **Data & Validation:** Match duration: integer seconds; Server secret: SHA-256 hash verified.

- **Business Rules:** BR-41 (Backend accepts results only from authorized server), BR-43 (Persistent data in PostgreSQL).
- **Normal Flow:** Server sends result; backend records transaction atomically; returns HTTP 200 OK.
- **Abnormal Flow:** Duplicate result submission for completed session returns HTTP 200 idempotently without double-counting stats.

ID	Requirement	Priority
FR-TEL-01	At match end, the authoritative dedicated game server shall report the final match outcome (Survivor Victory or Shade Victory), match duration, and per-player final status to the backend platform for persistent storage in PostgreSQL.	Must
FR-TEL-02	The system shall record per-match gameplay statistics (including assigned role, escape/downed status, and machine repair contributions) and update each player's aggregated career statistics in the database.	Should

3.10.4 Match Result Presentation

Function Trigger: Triggered on client when Dedicated Server broadcasts ClientRpc with final outcome.

Function Description:

- **Actors:** Player
- **Purpose:** Displays victory/defeat banner, match duration, individual performance metrics, and return to Main Menu button.
- **Interface:** Match Result Screen on Unity Client.
- **Data Processing:** Receives outcome enum, displays MSG-17 (Survivor Victory) or MSG-18 (Shade Victory), renders per-player summary table.

Screen Layout: [Placeholder: Insert Match Result Screen (Winner Banner & Performance Summary) Screenshot]

Function Details:

- **Data & Validation:** Display elements: Winner side, duration, role badge, status (Escaped/Downed/Killed).
- **Business Rules:** BR-35 (Single conclusive result display).
- **Normal Flow:** Clients receive outcome; transition to Match Result Screen; players review stats and click Return to Menu.
- **Abnormal Flow:** Disconnect during result screen returns player to Main Menu cleanly.

ID	Requirement	Priority
FR-TEL-03	The client Match Result Screen shall display the winning side, match duration, and a per-player performance summary at match end before returning players to the Main Menu.	Must

3.11 Cosmetic Skin Store & Payment

Enables browsing cosmetic character skins, purchasing via integrated Sandbox Payment Gateway, inventory unlocking, and equipping skins for matches (UC-15, UC-16).

3.11.1 Browse & Equip Cosmetic Skins

Function Trigger: Triggered when a Player opens the Skin Store Screen from the Main Menu.

Function Description:

- **Actors:** Player
- **Purpose:** Allows players to view the catalog of available character skins, check ownership, and equip owned skins for specific roles.
- **Interface:** Skin Store Screen on Unity Client via GET /api/v1/skins.
- **Data Processing:** Fetches available skins from Skin table and player inventory from UserSkin table; updates equipped status upon selection.

Screen Layout: [Placeholder: Insert Skin Store Catalog & Character Preview Screenshot]

Function Details:

- **Data & Validation:** Skin catalog: ID, name, character role, price, preview sprite.
- **Business Rules:** BR-49 (Skin ownership and equipment rules).
- **Normal Flow:** Player navigates to Store; browses skins; selects an owned skin and clicks Equip; equipped badge updates.
- **Abnormal Flow:** Attempting to equip an unowned skin prompts player to initiate purchase.

ID	Requirement	Priority
FR-STR-01	An authenticated Player shall be able to browse available character cosmetic skins, their prices, and ownership status in the Unity Client store.	Should
FR-STR-05	An authenticated Player shall be able to select and equip an owned cosmetic skin for use in matches.	Could

3.11.2 Skin Purchase & Payment Processing

Function Trigger: Triggered when a Player clicks Buy Skin on an unowned item in the Skin Store.

Function Description:

- **Actors:** Player
- **Purpose:** Creates a pending store transaction, processes payment with the Sandbox Payment Gateway, and unlocks skin upon verified completion.
- **Interface:** Payment Checkout Modal on Unity Client via POST /api/v1/store/purchase.
- **Data Processing:** Backend creates PENDING Transaction; requests payment checkout URL/token from Sandbox Gateway; upon callback, marks Transaction PAID and inserts UserSkin record.

Screen Layout: [Placeholder: Insert Sandbox Payment Gateway Checkout Modal Screenshot]

Function Details:

- **Data & Validation:** Price: currency amount; Transaction status: PENDING -> PAID or FAILED.
- **Business Rules:** BR-49 (Payment verification required before inventory unlock).
- **Normal Flow:** Player clicks Buy; checkout opens; player confirms mock payment; backend receives webhook; skin is unlocked (MSG-20); inventory refreshes.
- **Abnormal Flow:** Cancelled or failed payment marks transaction FAILED and displays MSG-26 without unlocking skin.

ID	Requirement	Priority
FR-STR-02	An authenticated Player shall be able to initiate a purchase for an unowned cosmetic skin, creating a pending transaction record on the backend.	Should
FR-STR-03	The system shall process skin purchase payments through the configured Payment Gateway (Sandbox/Mock integration) and verify transaction completion before granting skin ownership.	Should
FR-STR-04	Upon verified payment completion, the backend shall atomically record the completed transaction and unlock the cosmetic skin in the player's inventory.	Should

3.12 Bug & Issue Reporting

Provides players with an in-game reporting channel for bugs and gameplay glitches for administrator triage (UC-32).

3.12.1 Submit Bug & Issue Report

Function Trigger: Triggered when a Player opens the Report Bug modal from the Main Menu or pause menu, or when a Dedicated Server aborts.

Function Description:

- **Actors:** Player
- **Purpose:** Submits structured bug reports with description and client logs to backend for administrator triage.
- **Interface:** Bug Report Modal on Unity Client via POST /api/v1/reports.
- **Data Processing:** Validates input text, captures current session/build metadata, creates OPEN BugReport record in PostgreSQL.

Screen Layout: [Placeholder: Insert In-Game Bug & Issue Report Submission Modal Screenshot]

Function Details:

- **Data & Validation:** Category: GAMEPLAY, AUDIO, VISUAL, NETWORK, OTHER; Description: 10 to 1000 characters; Optional screenshot/logs.
- **Business Rules:** BR-43, BR-52.

- **Normal Flow:** Player encounters issue; opens modal; enters category and description; submits; backend saves report; client displays MSG-28.
- **Abnormal Flow:** Submitting empty description displays validation error. Network error prompts user to retry.

ID	Requirement	Priority
FR-TEL-04	The system shall allow players to submit bug or gameplay issue reports with optional diagnostic logs from the client to the backend for administrator review.	Should

3.13 Web Admin Portal

Dedicated browser-based administration console for player account moderation, versioned balance configuration tuning, match record auditing, bug triage, dashboards, and transaction reconciliation (UC-24, UC-25, UC-26, UC-27, UC-28, UC-29, UC-30, UC-31).

3.13.1 Administrator Authentication

Function Trigger: Triggered when an administrator accesses the Web Admin Portal URL.

Function Description:

- **Actors:** Admin
- **Purpose:** Authenticates administrative credentials and enforces ADMIN role privilege for all portal features.
- **Interface:** Admin Login Web Screen.
- **Data Processing:** Validates username/password, verifies ADMIN role authority, returns signed admin JWT token.

Screen Layout: [Placeholder: Insert Web Admin Portal Login Screen Screenshot]

Function Details:

- **Data & Validation:** Admin credentials: required; Role: must be ADMIN.
- **Business Rules:** BR-50 (Only authenticated users with ADMIN role may access Admin Portal).
- **Normal Flow:** Admin enters credentials; system verifies ADMIN role; redirects to Admin Dashboard.
- **Abnormal Flow:** Invalid credentials returns error. Non-admin player credentials receive HTTP 403 Forbidden.

ID	Requirement	Priority
FR-ADM-01	Administrators shall sign in to a dedicated React web portal that is separate from the Unity game client.	Must

3.13.2 Player Account Management

Function Trigger: Triggered when an Admin selects Player Management from the portal navigation.

Function Description:

- **Actors:** Admin
- **Purpose:** Allows searching, filtering, inspecting player profiles, and toggling account suspension or ban status.
- **Interface:** Player Management Web Screen via GET/PUT /api/v1/admin/users.
- **Data Processing:** Queries User table with pagination; updates enabled flag (true = ACTIVE, false = DEACTIVE); invalidates active sessions.

Screen Layout: [Placeholder: Insert Admin Player Management & Moderation Screen Screenshot]

Function Details:

- **Data & Validation:** Search query: username/email; Action: ACTIVATE (enabled = true) or DEACTIVATE (enabled = false).
- **Business Rules:** BR-50 (Admin governance over player accounts).
- **Normal Flow:** Admin searches player; clicks Ban; confirms action; user status updates; user is blocked from login.
- **Abnormal Flow:** Attempting to ban another ADMIN account is restricted.

ID	Requirement	Priority
FR-ADM-02	Administrators shall be able to search registered users, inspect player profiles and statistics, and manage account activation status (ACTIVE or DEACTIVE).	Must

3.13.3 Balance Configuration Management

Function Trigger: Triggered when an Admin selects Balance Configuration from the portal navigation.

Function Description:

- **Actors:** Admin
- **Purpose:** Inspects current gameplay tuning values and publishes updated versioned configuration sets applied to new match instances.
- **Interface:** Balance Configuration Web Screen via GET/POST /api/v1/admin/balance.
- **Data Processing:** Retrieves active BalanceConfig; validates numeric bounds; inserts new versioned record; Dedicated Servers fetch latest version.

Screen Layout: [Placeholder: Insert Balance Configuration Editor Screen Screenshot]

Function Details:

- **Data & Validation:** Tunables: repairTarget (1-10), sanityDrainRate (0.1-10.0), staminaCapacity (50-200), shadeDebuffMultiplier (0.1-1.0).
- **Business Rules:** BR-44 (Versioned gameplay balance configuration management).
- **Normal Flow:** Admin adjusts slider (e.g. required repairs from 3 to 4); clicks Publish New Version; new configuration activates for subsequent matches.
- **Abnormal Flow:** Submitting out-of-bounds numeric value displays validation alert.

ID	Requirement	Priority

FR-ADM-04	Administrators shall be able to view and edit versioned game balance configurations (such as required repair counts, Sanity rates, Stamina values and Shade light debuffs).	Must

3.13.4 View Match Records

Function Trigger: Triggered when an Admin selects Match Records from the portal navigation.

Function Description:

- **Actors:** Admin
- **Purpose:** Provides searchable and filterable historical audit of completed match sessions and active server instances.
- **Interface:** Match Records Web Screen via GET /api/v1/admin/matches.
- **Data Processing:** Queries GameSession and MatchPlayer tables with date, map, and outcome filters; displays detailed match timeline and participant roster.

Screen Layout: [Placeholder: Insert Match Records & Historical Session Audit Screen Screenshot]

Function Details:

- **Data & Validation:** Filters: date range, outcome (SurvivorVictory/ShadeVictory/Aborted), participant username.
- **Business Rules:** BR-43 (Persistent match session audit records).
- **Normal Flow:** Admin enters date filter; clicks a match row; inspects participants, roles, durations, and machine repair events.
- **Abnormal Flow:** No records found displays empty state placeholder.

ID	Requirement	Priority
FR-ADM-03	Administrators shall be able to view active rooms/dedicated servers and inspect persisted match history records. [Note: Running server monitoring to be reviewed with Dashboards in SDD].	Must

3.13.5 Review Bug & Issue Reports

Function Trigger: Triggered when an Admin selects Bug Reports from the portal navigation.

Function Description:

- **Actors:** Admin
- **Purpose:** Enables administrators to triage, review descriptions/logs, and update resolution status of player reports.
- **Interface:** Bug & Issue Reports Web Screen via GET/PUT /api/v1/admin/reports.
- **Data Processing:** Queries BugReport table; filters by severity and category; allows setting status (INVESTIGATING, RESOLVED, DISMISSED) with admin notes.

Screen Layout: [Placeholder: Insert Bug & Issue Reports Triage Screen Screenshot]

Function Details:

- **Data & Validation:** Status: OPEN, INVESTIGATING, RESOLVED, DISMISSED; Admin note: max 1000 characters.
- **Business Rules:** BR-43, BR-50, BR-52.
- **Normal Flow:** Admin clicks open report; reviews player description and server log snippet; marks status as RESOLVED.
- **Abnormal Flow:** Updating non-existent report returns error.

ID	Requirement	Priority
FR-ADM-05	Administrators shall be able to view, filter, and update the resolution status of player-submitted bug/game issue reports.	Must

3.13.6 Analytical Dashboards

Function Trigger: Triggered upon landing on the Admin Portal home page or selecting Dashboard.

Function Description:

- **Actors:** Admin
- **Purpose:** Renders high-level platform visual KPIs: concurrent sessions, match completion volume, and Survivor vs. Shade win rates.
- **Interface:** Admin Analytics Dashboard Web Screen via GET /api/v1/admin/dashboard.
- **Data Processing:** Aggregates match counts, computes side win rate percentages, active room counts, and daily active player metrics.

Screen Layout: [Placeholder: Insert Analytical Dashboard (Win Rate Charts & Platform KPIs) Screenshot]

Function Details:

- **Data & Validation:** Charts: Side win rate pie/donut chart, match volume bar chart, active room counter.
- **Business Rules:** BR-50 (Platform monitoring and analytics).
- **Normal Flow:** Admin opens dashboard; charts render in real-time displaying current balance indicators (e.g. 52% Survivor win rate).
- **Abnormal Flow:** Data aggregation failure displays cached figures with warning banner.

ID	Requirement	Priority
FR-ADM-06	Administrators shall be able to view an analytical dashboard displaying match volume, active sessions, and the win rate of each side (Survivors vs. The Shade).	Must

3.13.7 Store Transaction Auditing

Function Trigger: Triggered when an Admin selects Transactions from the portal navigation.

Function Description:

- **Actors:** Admin

- **Purpose:** Provides full audit trail of cosmetic skin purchases, payment provider references, and reconciliation statuses.
- **Interface:** Transaction Auditing Web Screen via GET /api/v1/admin/transactions.
- **Data Processing:** Queries Transaction table; displays transaction ID, timestamp, player username, skin name, amount, provider ref, and status (PAID/FAILED/PENDING).

Screen Layout: [Placeholder: Insert Store Transaction Auditing & Reconciliation Screen Screenshot]

Function Details:

- **Data & Validation:** Status filters: ALL, PAID, FAILED, PENDING.
- **Business Rules:** BR-49, BR-50 (Financial transaction audit and reconciliation).
- **Normal Flow:** Admin searches transaction by external reference; views full payment lifecycle timestamps.
- **Abnormal Flow:** No records found displays empty state indicator.

ID	Requirement	Priority
FR-ADM-07	Administrators shall be able to view, filter, and reconcile cosmetic skin store purchase transactions and payment statuses.	Must

4. Non-Functional Requirements

4.1 External Interfaces

Interface	Description
User interfaces	Unity 2D Game Client (1080p desktop UI): Login/Register screens, Main Menu, Room Lobby screen (6-character room code, 4-player readiness list, Host Kick/Start controls), Profile & Match History screen, Skin Store & Payment checkout screen, In-Game HUD (Health, Sanity, Stamina, role ability/drone view, shared machine repair counter, interaction prompt, on-screen Repair Minigame overlay, Reconnect/Bug Report modals), and Match Result screen. Web Admin Portal (responsive browser UI): Admin Login, Analytical Dashboard, Player Management, Balance Configuration editor, Match Records viewer, Bug & Issue Reports viewer, and Transaction Management table.
Hardware interfaces	Client: Windows PC with keyboard and mouse input, 1080p display, audio output, and an active broadband Internet connection. Server/Backend: Linux host / container runtime running headless Dedicated Game Server instances, Spring Boot Backend, and PostgreSQL.

Software interfaces	Game Client & Dedicated Server: Unity 6 (6000.6.x LTS, 2D URP), Unity Netcode for GameObjects (NGO 2.12), Unity Transport Package (UTP 2.6). Backend Platform: Java 21, Spring Boot 3.5, Spring Security (JWT/BCrypt), Spring Data JPA, Flyway, PostgreSQL. Admin Portal: React web application. External Services: Sandbox/Mock Payment Gateway API, Docker container runtime, GitHub Actions CI/CD.
Communication interfaces	1) Client <-> Backend & Admin Portal <-> Backend: HTTPS REST/JSON APIs secured with Bearer JWT authentication. 2) Dedicated Game Server <-> Backend: Internal HTTP/HTTPS REST callbacks (/validate-token, /heartbeat, /result) authenticated via per-match X-Server-Secret header. 3) Client <-> Dedicated Game Server: Authoritative UDP session traffic over Unity Transport (UTP) and Netcode for GameObjects (NGO).

4.2 Quality Attributes

The table below specifies the non-functional quality attributes across Performance, Network & Synchronization, Reliability & Recovery, Security, Compatibility, Maintainability, Usability, Testability, and Cost.

ID	Requirement	Priority
NFR-PERF-01	The Unity game client shall maintain >= 50 FPS at 1920x1080 resolution on the target baseline Windows PC hardware (quad-core CPU, integrated/entry discrete GPU).	Must
NFR-PERF-02	Backend REST API endpoints for authentication, room operations, and profile queries shall respond within typical interactive latency under capstone demonstration load [Quantitative threshold if required: Pending Confirmation].	Should
NFR-NET-01	The multiplayer match session shall operate over the Internet between four Unity clients and a headless Linux Dedicated Game Server using Unity Netcode for GameObjects (NGO) and Unity Transport (UTP).	Must
NFR-NET-02	The dedicated game server shall replicate authoritative state changes (Health, Sanity, machine repair count, door	Must

	state, exit state, and match outcome) consistently to all connected clients.	
NFR-REL-01	A disconnected player shall not corrupt the remaining match state; their character shall remain stationary in-game and support reconnection during the active match, while the Deadlock Safeguard ensures the match terminates cleanly if all connected Survivors have escaped or been downed.	Must
NFR-REL-02	Match result reporting from the dedicated server to the backend (/result) shall be idempotent so that network retries cannot duplicate match records or double-count player statistics.	Must
NFR-SEC-01	Gameplay-critical state shall be controlled exclusively by the authoritative dedicated game server.	Must
NFR-SEC-02	Clients shall not be able to directly modify machine repair state, Health, Sanity, Stamina, role assignment, Emergency Exit state, or match outcome.	Must
NFR-SEC-03	User passwords, single-use join tokens, and dedicated server secrets shall be stored as cryptographic hashes (BCrypt), and protected backend/admin endpoints shall enforce JWT authentication and role-based authorization.	Must
NFR-COMP-01	The Unity game client shall run on Windows 10 and Windows 11 (64-bit), and the dedicated game server shall run as a headless 64-bit Linux (and local Windows development) build.	Must
NFR-COMP-02	The React Web Admin Portal shall operate on modern desktop web browsers (Google Chrome, Microsoft Edge, Mozilla Firefox).	Should
NFR-MAINT-01	The codebase shall keep authoritative game rules and backend domain logic separate from UI presentation and scene rendering.	Should
NFR-USE-01	The game client and room lobby workflow shall be playable by a first-time tester with clear on-screen prompts and feedback messages.	Should

NFR-TEST-01	Automated unit and integration tests shall cover Sanity calculations, repair/objective validation, 3v1 role assignment, win conditions, authentication, join-token verification, and idempotent match result persistence.	Should
NFR-COST-01	Hosting infrastructure and external services (including Sandbox payment integration) shall operate within free usage tiers, student cloud credits, or team-hosted machines.	Should

5. Requirement Appendix

This appendix lists the consolidated business rules, common requirements, standardized application messages, and requirement-to-epic traceability matrix for THE LAST SIGNAL.

5.1 Business Rules

The table below defines all confirmed business rules (BR-01 through BR-50) governing authentication, rooms, 3v1 gameplay, server authority, store transactions, and administration.

ID	Category	Business Rule
BR-01	Match Composition	Each match shall contain exactly four players: three Survivors and one Shade.
BR-02	Role Assignment	The system shall randomly assign one player to The Shade role and the other three players to Survivor roles when the match starts.
BR-03	Role Assignment	Each of the three Survivor roles (Drone Operator, Engineer, Logistic) shall be assigned to at most one player per match (unique 1-1-1 Survivor distribution).
BR-04	Role Assignment	A player's assigned role shall remain unchanged for the duration of the match.
BR-05	Room Management	Each room shall have a unique 6-character room code that authenticated players can use to join the room.
BR-06	Room Management	A room shall not contain more than four players.
BR-07	Room Management	A player shall not occupy more than one player slot in the same room or be in multiple active rooms simultaneously.
BR-08	Room Management	A room shall not start a match until four players are connected, all four players have marked Ready, and the Host explicitly requests match initiation.

BR-09	Room Management	Only the authorized Host of a WAITING room shall be allowed to kick players or request match initiation.
BR-10	Room Management	A room shall not accept new players after its match has entered the STARTING or IN_GAME state.
BR-11	Room Management	The backend shall manage room state and coordinate dedicated game server allocation. The room Host shall not act as the authoritative gameplay server in production matches.
BR-12	Survivor Attributes	Each Survivor shall have Health, Sanity, and Stamina attributes.
BR-13	Survivor Attributes	A Survivor shall enter the Downed state when their Health reaches zero.
BR-14	Survivor Attributes	A living Survivor shall consume Stamina while sprinting, and Stamina recovery shall follow the configured gameplay rules when not sprinting.
BR-15	Survivor Roles	The Engineer shall repair broken machines faster than the other Survivor roles.
BR-16	Survivor Roles	The Engineer shall have greater maximum Stamina capacity than the other Survivor roles.
BR-17	Survivor Roles	The Drone Operator shall be able to operate a recon drone (steerable spotlight and motion radar) for remote reconnaissance, subject to configured restrictions.
BR-18	Survivor Roles	The Logistic shall be able to temporarily stun The Shade through its role-specific ability when the configured activation and cooldown conditions are satisfied.
BR-19	Survivor Roles	Each Survivor shall be able to use only the role-specific ability permitted by their assigned role.
BR-20	Machine Repair	All living Survivors shall be able to interact with and repair any unrepaired broken machine via the on-screen repair minigame, regardless of their assigned role.
BR-21	Machine Repair	A broken machine shall count toward the shared repair objective only after the authoritative dedicated game server confirms that its repair minigame has been successfully completed.
BR-22	Lighting	A successfully repaired machine shall cause the room containing that machine to become illuminated.

BR-23	Sanity	Applicable darkness shall decrease Survivor Sanity. The Shade shall not have a Sanity attribute.
BR-24	Sanity	When a Survivor's Sanity falls below the configured threshold, the system shall apply the corresponding configured gameplay debuffs.
BR-25	Lighting	Light from repaired machine rooms, the Safe Zone, and other permanently lit areas shall apply the configured debuff to The Shade while The Shade is exposed to that light.
BR-26	Survivor Attributes	Health, Sanity, and Stamina values shall remain clamped within their configured valid ranges.
BR-27	Map and Objectives	The map shall contain a predefined number of broken machines at match start.
BR-28	Emergency Exits	Both Emergency Exit rooms shall remain locked and inaccessible until the required number of machines (n) has been repaired.
BR-29	Map and Objectives	The Safe Zone shall be illuminated and shall serve as the Survivor spawn area at match start, while The Shade spawns at a separate dedicated Hunter spawn point.
BR-30	Map and Objectives	The map shall contain additional permanently illuminated areas besides the Safe Zone and repaired machine rooms.
BR-31	Escape	When Emergency Exits are unlocked, a non-downed Survivor entering an Emergency Exit zone shall immediately transition to the Escaped state and be removed from active map play while remaining active Survivors continue playing.
BR-32	Victory Conditions	The Survivor side shall win when at least one Survivor successfully escapes through an unlocked Emergency Exit, with the match concluding once all remaining active Survivors have either Escaped or been Downed.
BR-33	Victory Conditions	The Shade shall win when all three Survivors are Downed before any Survivor escapes.
BR-34	Match Rules	A match shall have no gameplay time limit.
BR-35	Match Rules	A match outcome shall be finalized and recorded exactly once when a valid win condition or session abort condition is satisfied.
BR-36	Game Server Authority	The dedicated game server shall be authoritative over gameplay-critical state, including role assignment

		enforcement, Health, Sanity, Stamina, machine repair progress, lighting effects, Emergency Exit availability, and match outcome.
BR-37	Game Server Authority	Clients shall request gameplay state changes from the authoritative dedicated game server and shall not directly determine authoritative match outcomes.
BR-38	Session Management	The backend shall provide dedicated game server connection information and join tokens to eligible room players only after the allocated server session is initialized.
BR-39	Session Management	The backend shall prevent repeated or concurrent match-start requests from creating multiple active game sessions for the same room.
BR-40	Session Management	The system shall handle room and dedicated server connection failures according to the configured session policy without accepting invalid room or match state transitions.
BR-41	Match Results	The backend shall accept a final match result only from an authorized dedicated game server presenting a valid server secret and shall prevent duplicate result processing.
BR-42	Runtime State	Gameplay runtime state shall be maintained in memory by the authoritative dedicated game server for the duration of the match.
BR-43	Data Persistence	Persistent user accounts, player career statistics, match session records, balance configurations, bug reports, skin ownership, and store transactions shall be stored by the backend in PostgreSQL.
BR-44	Configuration	Configurable gameplay balance values, including required repair count, Sanity rates/thresholds, Stamina parameters and Shade light debuffs, shall be managed through versioned configuration data.
BR-45	Authentication	Unauthenticated users (Guests) must register or log in before accessing the Main Menu, Room Lobby, Profile, or Store; valid stored JWT tokens enable automatic login on subsequent client launches.
BR-46	Room Management	If the Host leaves a room that is in the WAITING state, Host authority shall automatically transfer to the next remaining player in the room; if the last player leaves, the room shall be closed.

BR-47	Disconnect & Reconnect	When a player disconnects during an active match, their character shall remain stationary in-game and may be reclaimed upon reconnecting with their match session credentials; if all remaining connected Survivors have either Escaped or been Downed (or upon reconnect timeout), the server shall conclude the match to prevent deadlock.
BR-48	Downed & Revive	A Downed Survivor crawls at reduced speed (35% default), cannot repair machines or trigger escape alone, does not bleed out over time, and is restored to partial Health (35% default) when revived by a living teammate.
BR-49	Store & Payment	Cosmetic skin purchases require payment confirmation from the configured Payment Gateway before the backend unlocks the skin in the player's inventory and marks the transaction as PAID.
BR-50	Administration	Only authenticated users with the ADMIN role may access the Web Admin Portal to manage players, balance configurations, match records, bug reports, dashboards, and store transactions.
BR-51	Machine Repair	If a machine repair minigame fails or is cancelled by Survivor movement or interruption, the repair progress is not completed and the machine remains broken and immediately retriable.
BR-52	Bug & Issue Reporting	An authenticated Player may submit a bug or gameplay issue report containing a category, title, description, and client telemetry logs; reports are assigned an initial OPEN status for administrator review.

5.2 Common Requirements

All user interface text across the Unity Game Client and React Web Admin Portal is in English.

The game client targets Windows 10/11 (64-bit) PC with keyboard and mouse input, while the dedicated game server runs as a headless Linux build (LI-01). Controller, console, and mobile platforms are out of scope.

Player accounts, career statistics, match session records, cosmetic skin inventories, store transactions, versioned balance configurations, and bug reports are persisted by the Spring Boot backend in PostgreSQL.

Gameplay runtime state is maintained authoritatively by the dedicated game server for the duration of each match.

Offline single-player practice mode, global leaderboards, and public matchmaking queues are out of scope for the capstone release.

5.3 Application Messages List

The table below defines the standard user-facing messages displayed across the Unity Game Client and Web Admin Portal following the Report 3 message catalog format.

#	Message Type	Context	Content
MSG-01	Inline / Modal	User submits login form with invalid username or password	Incorrect username or password. Please check again.
MSG-02	Toast / Modal	Guest completes account registration successfully	Account registered successfully.
MSG-03	Status Text	Host initiates room creation from Main Menu	Creating room...
MSG-04	Toast / Status	Backend confirms new WAITING room creation	Room created.
MSG-05	Status Text	Player submits 6-character code to join a room	Joining room...
MSG-06	Inline / Modal	Room code is invalid, closed, already in-game, or full (4/4)	Failed to join: room code is invalid, unavailable, or full.
MSG-07	Status Text	Room lobby has fewer than four connected players	Waiting for players...
MSG-08	Toast / Banner	Host leaves a WAITING room and Host authority transfers	Host left the room. Host authority has been transferred.
MSG-09	Modal	Host kicks a non-host player from a WAITING room	You have been removed from the room by the Host.
MSG-10	Status Text	All four players are Ready and Host initiates match start	Match starting... Allocating dedicated server.
MSG-11	HUD Banner	Dedicated server completes random 3v1 role assignment at match start	Roles assigned.
MSG-12	HUD Notification	A Survivor successfully completes a machine repair minigame	Machine repaired. Room lighting restored.

MSG-13	HUD Warning	A Survivor fails or cancels an active repair minigame	Repair interrupted! Machine not yet operational.
MSG-14	HUD Banner	Required number of machines (n) has been repaired	Emergency exits unlocked! Proceed to an extraction zone.
MSG-15	HUD Alert	A Survivor reaches zero Health and enters the Downed state	Survivor downed! A teammate must revive them.
MSG-16	HUD Banner	A living Survivor enters an unlocked Emergency Exit	Survivor escaped!
MSG-17	Result Banner	Match ends with at least one Survivor escaped	Survivor Victory - Survivors escaped!
MSG-18	Result Banner	Match ends with all three Survivors downed and zero escapes	Shade Victory - The Shade wins!
MSG-19	Overlay / Banner	Player loses connection to the Dedicated Server during an active match	Connection lost. Attempting to reconnect to match...
MSG-20	Toast / Modal	Payment gateway callback completes during Skin Store purchase	Payment processed. Cosmetic skin unlocked in your inventory.
MSG-21	Inline / Modal	User attempts to register with an existing username or email	Username or email already in use. Please choose another.
MSG-22	Toast / Modal	Player successfully changes account password	Password updated successfully.
MSG-23	Inline / Modal	Player enters incorrect current password during password change	Current password is incorrect.
MSG-24	Modal	Deactivated user (enabled = false) attempts to authenticate	Your account has been deactivated. Please contact administration.
MSG-25	Modal / Toast	Expired or invalid authentication token encountered	Session expired. Please log in again.
MSG-26	Toast / Modal	Payment transaction is cancelled, times out, or fails verification	Payment could not be completed. Transaction cancelled.

MSG-27	Toast / Banner	Host attempts to start match when not all 4 players are Ready	Cannot start match: all players must be Ready.
MSG-28	Toast / Modal	Player successfully submits a bug or issue report	Report submitted successfully. Thank you for your feedback.

5.4 Other Requirements

The table below traces each functional and non-functional requirement group in Section III back to the project's task packages (Epics) defined in Section II.1.1.

Requirement Group	Related Task Package (Epic)
3.2 Account & Profile Management (FR-AUTH-*, FR-PRO-*)	Epic 5 - Backend Platform (accounts, profiles, results)
3.3 Room & Lobby Management (FR-NET-01..04, FR-NET-10..13)	Epic 3 - Network Infrastructure; Epic 9 - Rooms & Lobby
3.4 Match Session & Synchronization (FR-NET-05..09, FR-NET-14..15)	Epic 3 - Network Infrastructure; Epic 4 - Dedicated Server & Orchestration
3.5 Core Movement, Interaction & Map Exploration (FR-CORE-*, FR-MAP-01,04,05,09,10, FR-OBJ-01)	Epic 2 - Core Gameplay Prototype; Epic 8 - Repairs, Objectives & HUD
3.6 Survivor Survival & Abilities (FR-SUR-01..08, FR-SUR-10)	Epic 6 - Character Roles, Survival & Toolsets
3.7 The Shade - Hunter Gameplay (FR-HUN-01..06)	Epic 6 - Character Roles, Survival & Toolsets
3.8 Machine Repair & Exit Objectives (FR-PUZ-*, FR-MAP-02,07, FR-SUR-09, FR-OBJ-02,03)	Epic 7 - Sanity & Interactive Minigames; Epic 8 - Repairs, Objectives & HUD
3.9 Darkness, Lighting & Sanity (FR-MAP-03,06,08, FR-SAN-01..06)	Epic 7 - Sanity & Interactive Minigames; Epic 8 - Repairs, Objectives & HUD
3.10 Match End & Results Processing (FR-OBJ-04..07, FR-TEL-01..03)	Epic 5 - Backend Platform; Epic 8 - Repairs, Objectives & HUD; Epic 12 - System Integration
3.11 Cosmetic Skin Store & Payment (FR-STR-01..05)	Epic 5 - Backend Platform; Epic 10 - Admin Portal & Moderation
3.12 Bug & Issue Reporting (FR-TEL-04)	Epic 10 - Admin Portal & Moderation; Epic 11 - Telemetry & Balance Dashboard

3.13 Web Admin Portal Management (FR-ADM-01..07)	Epic 10 - Admin Portal & Moderation; Epic 11 - Telemetry & Balance Dashboard
4.2 Quality Attributes (NFR-PERF-*, NFR-NET-*, NFR-REL-*, NFR-SEC-*, NFR-COMP-*, NFR-TEST-*)	Epic 12 - System Integration; Epic 13 - Beta Testing; Epic 14 - Release